Math 512 Data Science Project Version 4

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1 Introduction

With the recent releases of *Barbie* and *Oppenheimer*, more attention than ever has been placed on movies and their potential to generate enormous profits. If the formula for producing the next blockbuster was effortless, production companies would be first in line to churn out as many of them as possible. However, the formula for success is far from simple. Over the past several years there have been a plethora of failed ventures in which studios lost millions of dollars. One can look no further than the recent DC studio film, *The Flash*, which saw $268,533,313 in domestic and international revenue generated at the box office [15]. Due to a production budget of $200,000,000 and an equally large budget dedicated to marketing, the revenue did not substantially offset the investment. As a result, DC studios slated the film as a flop and was inevitably a failed expenditure. With this in mind, it raised several questions. What factors go into creating a box office success? Is it even possible to predict the box office to begin with? What is the best way to approach this problem?

Throughout this project, we are going to take on the role of movie studio consultants. Our job is to answer these exact questions. To do so, we want to have a concrete methodology that can predict the risk of investment and any likely outcomes. We plan on taking into consideration a variety of factors, including but not limited to, the production budget of the movie, the advertising budget, the lead actors, the date that the movie is released, any relevant social media/word-of-mouth buzz, and Rotten Tomato Reviews. In light of all of this, we seek to implement the tools of data science to construct an algorithm that will most accurately estimate a movie's domestic gross, international gross, and opening weekend gross. Additionally, we aim to assess how gross revenue changes over the life cycle of the movie alongside the importance of the release date. Equally as important, we are going to prioritize analysis regarding the Chinese box office to monitor how dependent international box office success is on China's box office gross revenue. To manage this mountain of a task, we are going to apply several machine-learning techniques in Python. These will include Miss Forest (MF), Multi-Linear Regression (MLR), K-nearest neighbors (KNN), Regression, Logistic Regression, and more. Through the implementation of all of these different machine learning techniques, we seek to determine which one is the strongest at predicting the success of movies at the box office utilizing area under the ROC Curve (AUC) scores. This score looks at the area under a receiver operating characteristic curve (ROC curve) and is a commonly used tool to ascertain the predictive power of an algorithm. A score of 0.9/1.0 is often used as an industry standard.

2 Literature Review

Our group is far from the first to attempt to predict the box office, nor will we be the last. Thus it is important to look at past projects by other collaborators in the field of data science to give us a look ahead at what we can expect to see throughout our project.

The first part of the elaborate process typically begins with attaining and pre-processing data. We can see some colleagues have scraped data from text-heavy sources like *Box Office Mojo* and *Wikipedia*. [18]. Others used database-heavy sites such as *SQLite* and *IMDbPY* [6], to *Kaggle* to achieve similar results [7]. Another group focused on data collection using movie-specific sites like *IMDb* and *TMDb*, from which a lot of data sets present on other websites typically derive their information anyways [16]. On the other hand, international studies commonly used sources such as *Maoyan* [9], *Weibo*, and the Chinese Box Office Revenue Website to siphon non-domestically centered data [10]. *Maoyan*, for example, was used frequently to collect ticketing information for these studies. After collecting data from various sources, pre-processing is the next key step as it removes incomplete or duplicated items which can unnecessarily skew results. [6] used SQL queries to omit rows that weren't movies, rows that were adult films, rows lacking a set budget range, and rows that had incomplete data. Likewise, a separate team eliminated all rows with missing values and duplicates [2]. Upon cleaning the data as necessary, this team looked into splitting genres of movies with multiple genres and converted categorical columns into numerical values as necessary via dummy values. A separate team [13] did the same, using dummy values for its categorical features, and went even further by using an auto-encoder neural network trained with unsupervised learning to extract latent variables. This was followed by the use of K-Means clustering to cluster the reduced data set. The silver lining from all

these previous studies shows that pre-processing and data collection are the critical steps needed before any modeling can even start.

While data collection is inherently obvious in creating a prediction system, the question remains how much data is enough, moreover, in our project, we would like to gauge how large the data set should be. To get a good idea, we can see that on one end of the spectrum, a 2020 study amassed a large data set of 35,794 movies from the period of 1920-2020 [4]. On the other hand, studies such as [14] and [16] looked at a smaller data set size, totaling only 152 domestic commercial films from 2018 and 289 movies from 2018-2019 respectively. Mid-sized data sets were more commonly employed as indicated by studies such as, [16], using a total of 3100 movies after pre-processing. Likewise, the team behind the study [8] looked at a total of 3167 movies from 1980 to 2019. The most diverse data set we saw covered 5043 movies over 100 years of release from 66 different countries [1]. Although it is important to have a data set rich in information, it is equally as important that information is relevant. The researchers behind [8] found that limiting data to smaller periods allowed for better alignment with any sudden cultural or technological shocks. Maintaining the importance of narrowing the window of time while still having a sizable enough amount of data to work with, these researchers found great success in revisiting their previous study with this framework in mind. This team also normalized their feature values as needed while using monetary profits free of inflation instead of box office revenue as its predictor. To do this, they used the 2019 Consumer Price Index. While many studies assumed the necessity of large sets, the researchers behind [11] took it a step further, utilizing Kaiser-Meyer-Olkin (KM0) sampling to ensure that their sample was indeed large enough. When looking at standardization and scaling, [2] found that feature scaling was necessary in their models. From these various studies, it is clear that there is no one-size-fits-all data set that will intrinsically perform better than another. However, as both studies [11] and [2] assert with their use of KMO testing, there are techniques in place to check if the data size collected is sufficient for data analysis. If insufficient, the used data could lead to inaccurate predictions.

After pre-processing is used to clean a data set, it becomes extremely important to identify what features one will enlist in a predictive model. Study [18], for example, collected many features such as premiere date, production name, budget, domestic gross, international gross, total gross, trailer views, studio, genre, and others. The authors in [13] looked at the movie name, month of release, production country, genre, director, actor, and screening time. The group behind [11] particularly incorporated release time using the "Film 360 Index", an index that records the search volume for a movie, on its release date as well as two weeks before its release. The authors of [14] looked at other holistic factors such as price, director, actor, and whether it was part of a sequence and genre. The researchers in [17], on the other hand, concentrated on more distinctive factors such as MPAA rating, competition level among movies (high, medium, low), the star value of actors, and more. Interestingly, from their results, the group concluded that star value was not a largely important predictive feature of box-office outcomes. Study [19] took a different approach altogether, reducing the factors to consider creativity, distribution, and marketing. They included the movie awards given to actors and directors from professional film organizations within their analysis of the three primary factors. Instead of focusing on revenue as many of the other studies had, this paper measured a movie's success based on critical acclaim within the industry. Thus instead of strictly monetary factors, elements related to industry approval were selected to predict success. When looking at which specific features had the best performance in predicting movie success, [2] found that actor, actress, and antagonist were the key elements that resulted in their model's prediction accuracy increasing the most. Their analysis suggests the choice of cast matters. [14] found that film directors, actors, and online word-of-mouth had a high influence on box office revenue, while movies with popular IPs could see up to a 5 percent increase in revenue. Interestingly, they also determined that the price of tickets and the release date of the film had little to do with the amount of money a film earned. The team behind [4], on the other hand, found that features such as actors and directors were hard to work with since, by machine-learning standards, there wasn't enough data for the movies that they were involved in. It is important to note that neural networks, which were the groups' modeling strategy, typically utilize a significant amount of information. Thus when there was little information regarding those factors specifically, the model inherently underperformed. Sifting through their analysis, the team behind [9] found mixed results between opening weekend sales and subsequent movie attendance. While the previous studies they cited in their literature review claimed a negative correlation between the two, the correlation they found was not strong enough to indicate that was necessarily the case. While piracy and a fixed audience

pool could decrease potential subsequent sales after a successful opening weekend, the effect they measured had "mixed signs" [9]. To definitively see the highest performing factor, one team employed the technique of assigning weights to attributes [1]. Humorously, they concluded that a film's success couldn't be predicted by relying solely on a single attribute.

Despite the importance of considering a multitude of factors, many studies suggest that word-of-mouth is among the most important when it comes to determining the potential box office revenue of a movie. This is something that also is visible before officially reviewing the data. The authors in [14] found that data collected from pre-production regarding word-of-mouth had pronounced spillover effects at the box office. In [9], the researchers looked at owned social media (OSM) which covers officially sponsored media and advertisements, as well as earned social media (ESM) which covers social media content from critics, influencers, fans, and people not officially related to the movie. This group found that OSM had a larger long-term impact than ESM, but reasoned OSM's strictly positive and scheduled nature created a long-lasting effect. On the other hand, ESM content was found to be typically sporadic whether positive or negative. This helped them qualitatively explain why ESM content did not last nearly as long. Similarly, [3] split its data into User-Generated content and Marketer-Generated content. They used a comprehensive analysis of *Facebook* and *Twitter* along with User-Generated Content and Marketer-Generated Content to create their model. The researchers in [5] emphasized the importance of the quality reviews that a movie receives when determining box office outcomes. In particular, from a review of their data, they hypothesized that movies with higher review helpfulness possessed higher predictive power for electronic word of mouth on current box office revenue levels. This source also looks at tweet count pre and post-release of a movie using *Textblob* to analyze the sentiments of tweets. One team that wasn't able to include social media buzz in their study was [7] however, they equally believed including data from *Twitter* and *YouTube* could help improve the predictive capabilities of their model despite their already high accuracy rate within their concluding remarks. If there is one factor that we must include in our project, it can certainly be argued to be this one.

In the inclusion of various factors, it is also valuable to utilize methods that can identify correlations between them. One such popular example is the use of the Pearson Correlation Coefficient. The authors of [11] incorporated this exact technique before using multicollinearity analysis to remove features from their model with too high multicollinearity. Another team [7] used Pearson or product-moment correlations to examine relationships among variables in a similar fashion.

We can see the Pearson Correlation Coefficient's Formula below (1):

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{1}$$

Machine Learning Techniques are the basis of Data science and what many teams before us used in their attempts to create the best predictive models possible. The team behind [18] utilized a shotgun approach, applying multiple techniques ranging from Multi-Linear Regression, Support Vector Machine Networks (SVM), regularized linear regression, K-nearest neighbors (KNN), decision trees, bagged trees, random forest, and more, in an attempt to determine which technique worked the best. On the other hand, the authors of [11] strictly applied regression analysis to obtain a linear equation for their predictor. Similarly, [14] implemented both the Litman model of OLS (Ordinary Least Squares) regression alongside the quantile regression which they used because of the results attained once their thresholds were utilized. This allowed the group to create their model and achieve an even greater comprehensive analysis. In their paper, [2], the researchers used SVM's, KNN modeling, Decision Trees, and Random Forest modeling in addition to logistic regression. Compare that to the authors of [12], which made use of Multiple regression analyses to calculate R-squared values accompanied by a 10-tree Random Forest with Decision Tree Boosting. They also employed KNN fitting with K = 1 to help find patterns and relationships that the other models could not. In other studies, less popular techniques such as polynomial regression models or the Naive Bayes method were implemented in [5] and [6] respectively. The most popular ones, however, included the Random Forest models, such as in [17], as well as Ada Boost, Gradient Boost, and K-nearest neighbor models in [7]. Interestingly, one team even employed data mining techniques in creating their predictive system [7] while another used Panel Vector Auto Regressions as its sole approach.

Neural networks were also widely used by our previous peers. In particular, the researchers from [17] used support vector machines and a Multi-Layer Perceptron neural network for their predictions. Impressively as

described in [18], the authors created their own MLM(Masked Language Model) a self-supervised pre-trained network that wound up outperforming any other model they compared it against. This feat was achieved by pre-training the model on data that pruned low-budget films and any information that was not publicly available. While this reduced the potential set of data, it increased their accuracy immensely. On the other hand, one team used neural network implementation via *Neurosolutions* software rather than building it separately [17]. Two additional specific networks were used by our peers that we also noted as particularly intriguing. The first one is a Deep Belief Network [19] that is widely used in image classification and speech recognition as well as a Convolutional Neural Network [16]. Although neural networks can be very good machine learning models, the team behind the study [19] cites neural networks as being poor predictors because of their simplistic nature. This is something that we will keep in mind for our project development.

When finally looking at validation, several teams [2], [8], and [6] looked at the accuracy, precision, recall, and F-score of their models to determine the best one. One additional model utilized 10-fold cross-validation [17] to ensure the credibility of their model. This process divides the data set into ten mutually exclusive subsets as each fold is used to test the classifier's performance trained on the combined data of the remaining nine folds. This team also introduced Accuracy Percentage Hit Rate (APHR) which takes the number of samples correctly classified, C, divided by the total number of samples, S, to measure model accuracy and effectiveness. This is written as

$$R = \frac{C}{S}, \tag{2}$$

where R is the APHR mentioned above.

3 Data Sets Collected

For our project, we worked with three separate datasets. The first dataset was sourced from the *Kaggle* and contained information for 10,760 movies released between 1961 to 2015. This moderately sized data set aligned with what was most commonly used by previous researchers. All of the movies for this set were siphoned directly from *TMDb*, with the CSV file including a variety of attributes for each film. Specifically, these attributes were:

- Budget
- Revenue
- Title
- Cast
- Director
- Keywords associated with each movie
- Runtime in minutes
- Genre
- Production Company
- Release Date
- Popularity Level
- Popularity Score
- Adjusted Values of budget & revenue to 2010 Money valuations
- Profit
- *IMDb* identification

- Movie Website
- & More

The second dataset was the one provided to us at the beginning of the project, containing details from 619 movies between 2000-2019. The file also contained a plethora of factors for each film. This included items such as:

- Title of the film
- Domestic and International Box Office Sales
- Opening Weekend Revenue
- Release Date
- Production Cost
- Genre
- Rotten Tomatoes Rating (RT)
- Google Trends Score (GT)
- Ratios for International Sales-budget and Opening Weekend Revenue-Budget

Interestingly, the file also incorporated binary classifications for each film on whether or not it was a sequel, franchise, action, drama, comedy, adventure, summer, or holiday film. The film *Mission Impossible 3*, for instance, is given a 1 in the sequel, franchise, action, and summer categories, with a 0 for the others.

The purpose of using the first two data sets is to see how our models perform from one set to another. Likewise, we can analyze which factors were most important among the two in determining revenue. However, since the *Kaggle* data set does not contain opening weekend revenue values, we are only going to use the provided data set when trying to estimate opening weekend revenue.

The third data set that we are implementing is one that we have curated using a scraper of our own. Pulled from *the-numbers.com*, this data set contains 2107 movies released between 2013-2023 in which the movies were roughly within the top 100 international box office revenues for a given year. Its features include:

- Movie Title
- Total International Box office Revenue (all non-American countries)
- The name of the top non-U.S. market for each movie
- Box Office Revenue of the top Non-USA country
- Share of the number one market of the top non-U.S. country

The purpose of this set of data is to answer the question of whether or not movies need to do well in the Chinese market to do well internationally. This set excludes revenue from the U.S. box office to look strictly at international performance.

3.1 Cleaning and Pre-processing

As expected, all three data sets required both cleaning and pre-processing. Using the dropna() and drop() functions from the *Pandas Python* library, we removed any instances of missing values within each data set as well as the columns that were not of use. For example in the *Kaggle* data set, fields such as the movie websites, *IMDB* ID, taglines, and more were removed. Similarly in the Math512 set, duplicated and blank columns were immediately discarded. With the three data sets containing only the numerical, categorical, and monetary features we wanted to analyze, the next thing we had to address was the formatting.

For the *Kaggle* data set, features such as the cast, genre, and production company formatted the cell elements in such a way that the separate string values were all concatenated together by a "—" character. For example, the movie *Jurassic World* has a cast element of "Chris Pratt—Bryce Dallas Howard—Irrfan Khan—Vincent D'Onofrio—Nick Robinson" and a genre element of "Action—Adventure—Science Fiction—Thriller". Processing the data set, we split the strings to separate each distinct entry. This was accomplished using a "str().split()" function and indexing it three times to isolate the first three entries for each. The first index was assigned to a primary column, the second to a secondary column, and the third index was assigned to a tertiary column for each separate feature. A similar approach was applied to the release date string column, separating it into columns of release day and release month to mirror these same features already present in the Math512 data set.

In the Math512 data set, we found that the monetary-based columns formatted the numerical values as strings. Since dollar signs and commas were frequently concatenated within these strings, we removed them by creating a "cleanString()" function. This function takes in a string and removes all non-numerical values such as dollar signs, commas, or quotes and turns the string into a workable integer value. We then utilized the Pandas method: ".apply()" to apply this function to all the strings in a given feature column. This function was also useful in *the-numbers* data set, allowing us to remove the percent sign character present in the "Share of the Number One Market" feature. At this point, the three data sets were still a mix of numerical columns and string columns. However, with regression and prediction, we only want to work with numerical values. Thus we had to convert any categorical features into quantifiable ones.

To address this we created a generalized "IntegerAssign()" function that gave each unique entry in a categorical feature column or categorical feature column array (e.g. [PrimaryGenre, SecondaryGenre, ThirdGenre]) its non-ordinal numerical value. The ordinality means there is no correlation between numbers such as 1 being closer to 2 than to 50. This function was applied to features such as the genres, actors, and popularity levels in the *Kaggle* data set, the genre in the Math512 set, and the name of the top non-U.S. market feature in *the-numbers* data set. The *Kaggle* set's popularity level feature, for instance, rated movies on the scaling of "Low, Moderately Low, Moderately High, and High". These were transformed into numerical ratings of 0, 1, 2, and 3 respectively via the "IntegerAssign()" function. The complete mappings are provided in the first section of the appendix.

After implementing the mappings, we also wanted to incorporate features that we knew would be useful for achieving our project goals. For the *Kaggle* and Math512 data sets, a combined release number (CR) was introduced based on the formula

$$CR = x + y/31. \tag{3}$$

In this case, x is the integer (int) conversion of the string release month, and y is the int conversion of the release day. Thus, for strings "6" and "9", corresponding to June 9th, the CR would be 6.29 in either set. For the Math512 set, a total revenue feature was created to emulate the *Kaggle* set's revenue column, summing the international and domestic revenue values. For both the *Kaggle* and Math512 sets, a "Revenue to Budget" feature (Rev/Bud) was also constructed, being the ratio of revenue to budget features in a data set. Finally, for easier reading and analysis, we divided the monetary columns across all data sets by 1 million to get our values in terms of millions of dollars instead of in terms of dollars.

With this completed, the data sets, as shown in Figures 1, 2, and 3 were entirely numerical and ready for analysis.

	popularity	budget	revenue	director	runtime	vote_count	vote_average	budget_adj	revenue_adj	profit	...	PrimaryGenre	SecondaryGenre	ThirdGenre	PrimaryProd	SecondaryProd	ThirdProd	ReleaseMonth	ReleaseDay	CombinedRelease	Rev/Bud
0	32.986	150.0	1513.529	0	124	5562	6.5	138.0	1392.446	1363.529	...	0	1	3	0	279	47	6	9	6.290	10.090
1	28.420	150.0	378.436	1	120	6185	7.1	138.0	348.161	228.436	...	0	1	3	1	397	408	5	13	5.419	2.523
2	13.113	110.0	295.238	2	119	2480	6.3	101.2	271.619	185.238	...	1	3	10	2	398	452	3	18	3.581	2.684
3	11.173	200.0	2068.178	3	136	5292	7.5	184.0	1902.723	1868.178	...	0	1	3	3	399	149	12	15	12.484	10.341
4	9.335	190.0	1506.249	4	137	2947	7.3	174.8	1385.749	1316.249	...	0	7	10	4	237	84	4	1	4.032	7.928
5	9.111	135.0	532.951	5	156	3929	7.2	124.2	490.314	397.951	...	2	4	1	5	228	800	12	25	12.806	3.948
6	8.654	155.0	440.604	6	125	2598	5.8	142.6	405.355	285.604	...	3	0	10	6	400	408	6	23	6.742	2.843

Figure 1: Cleaned Snippet of the *Kaggle* data set

Title	Domestic Box Office	International Box Office	Production Cost	Genre	RT Rating	Opening Weekend Revenue	Sequel	Franchise	Action	Drama	...	Holiday	GT Score	INTER/BUDGET	OPEN/BUDGET	DOM/BUDGET	ReleaseMonth	ReleaseDay	CombinedRelease	Revenue	Rev/Bud
Pirates of the Carib: Dead Man	423.314	642.900	225.0	0	0.30	135.635	1	1	0	0	...	0	3.0	2.857	0.603	1.881	7	7	7.226	1066.214	4.739
Night at the museum	250.863	328.583	110.0	0	0.42	30.434	0	0	0	0	...	1	4.0	2.987	0.277	2.281	12	22	12.710	579.446	5.268
Cars	244.083	217.568	70.0	0	0.75	60.434	0	0	0	0	...	0	15.0	3.108	0.863	3.487	6	9	6.290	461.651	6.595
X-men Last Stand	234.362	224.898	210.0	1	0.57	102.751	1	1	1	0	...	0	39.0	1.071	0.489	1.116	5	26	5.839	459.261	2.187
The Da Vinci Code	217.536	550.284	125.0	2	0.25	77.073	0	0	0	0	...	0	18.0	4.402	0.617	1.740	5	19	5.613	767.820	6.143
Superman Returns	200.120	173.965	232.0	0	0.75	52.535	1	1	0	0	...	0	15.0	0.750	0.226	0.863	6	28	6.903	374.085	1.612
Happy Feet	198.000	187.000	85.0	0	0.76	41.533	0	0	0	0	...	1	16.0	2.200	0.489	2.329	11	17	11.548	385.000	4.529

Figure 2: Cleaned Snippet of the Math512 data set

	International Box Office Revenue	Box Office Revenue of Top Country	Share Of Number One Market	Number One Market	Remaining International Box Office Revenue
0	804.405	118.817	14.8	0	685.588
1	788.028	101.918	12.9	1	686.110
2	673.597	673.290	100.0	2	0.307
3	627.937	74.294	11.8	0	553.643
4	599.437	596.850	99.6	2	2.587
5	568.454	135.181	23.8	2	433.273
6	540.457	539.479	99.8	2	0.978
7	486.473	86.931	17.9	2	399.542
8	485.333	484.263	99.8	2	1.071
9	394.191	47.735	12.1	2	346.456

Figure 3: Cleaned Snippet of *the-numbers* data set

3.2 Correlation Matrices

As mentioned in our goals, we want to analyze the Pearson coefficients between features and identify any multi-collinearity between the many factors. The presence of multi-collinearity would cause our model's predictive ability to decrease significantly. Likewise, it would affect our interpretability of the results, where the effect of one factor could be overestimated due to its dependence on another. Applying the .corr() function included in the *Pandas* library for *Python*, we were able to calculate the necessary Pearson coefficients for the *Kaggle* and Math512 data sets. The .corr() function utilizes eq.(1) for each factor. Graphically, the *pandas* library allows all the calculated correlations to be formatted neatly into one large matrix, as discussed in our literature review. Using the symmetric threshold of "-0.85 - 0.85" for the coefficients and removing non-ordinal factors, we produced figures 4 and 5.

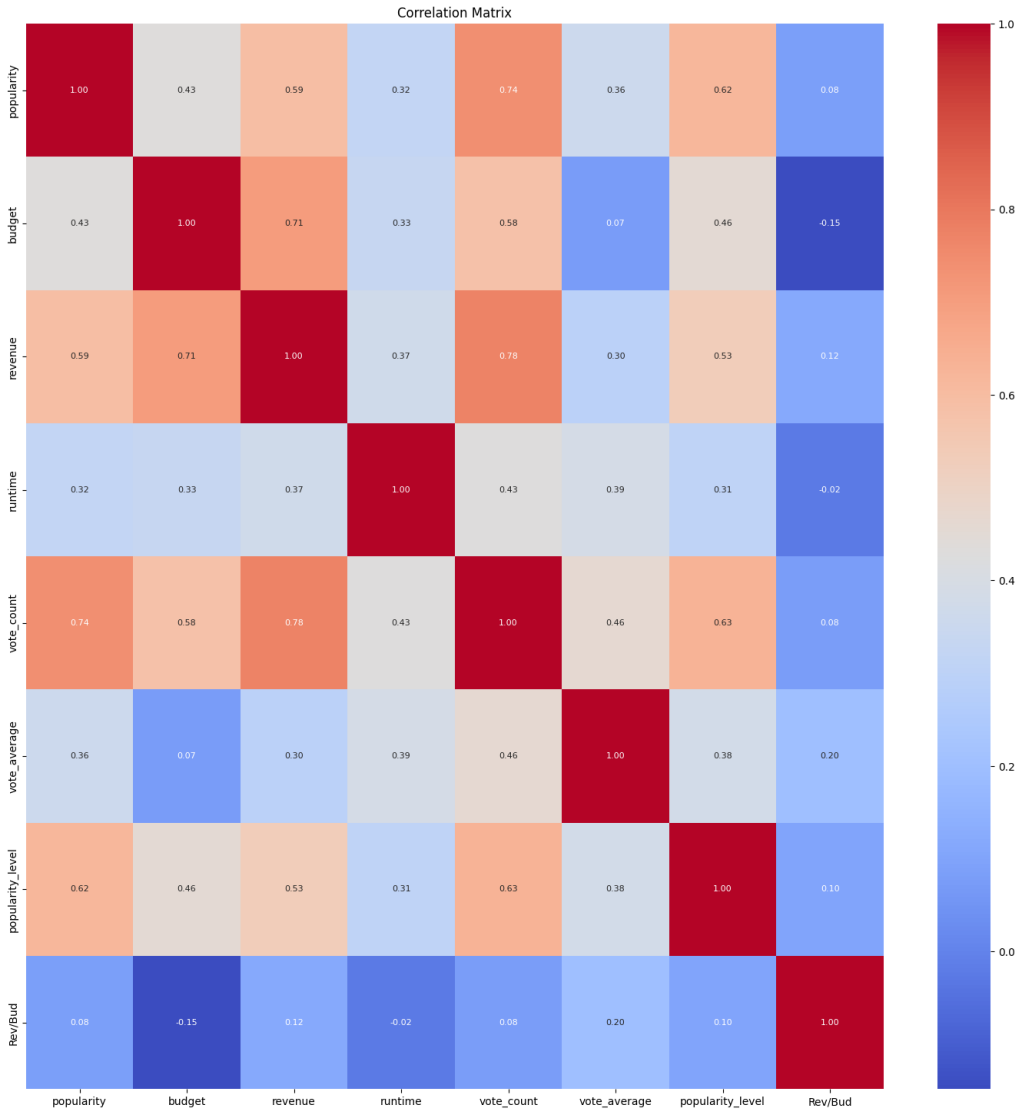

Figure 4: Correlation Matrix for *Kaggle* data set

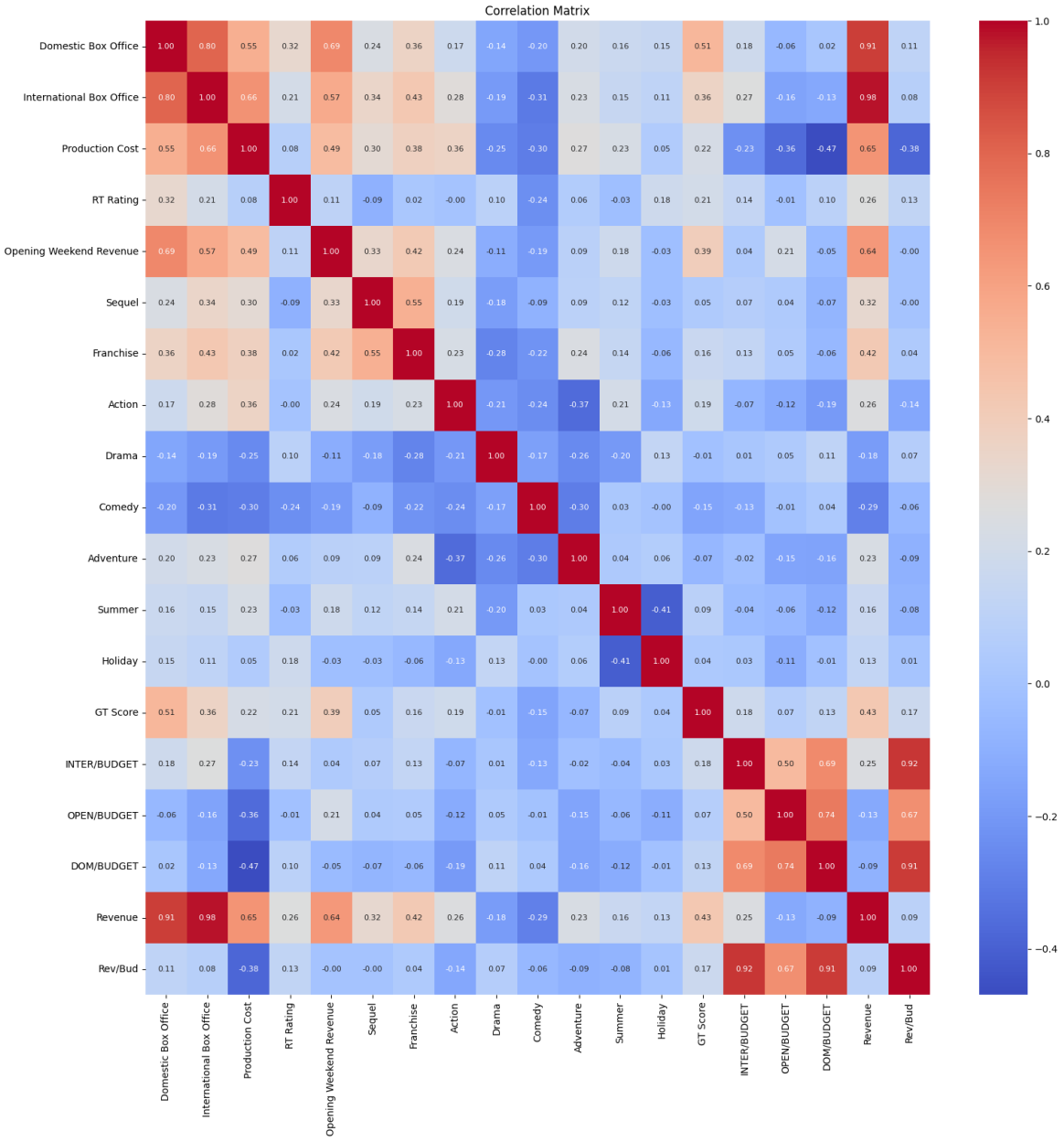

Figure 5: Correlation Matrix for Math512 data set

For the *Kaggle* data set, the factors of adjusted budget and adjusted revenue as well as release month and release day exceeded our threshold of large collinearity. Thus, these columns were then removed to avoid skewed results and try to create a better predictive model. Similarly, this was also found to be the case for profit and revenue. Since we want to treat revenue as the target, the profit column was removed as well.

Compared to the *Kaggle* data set, the matrix in Figure 5 indicated no overtly large multicollinearity that exceeded the previously defined threshold. This meant that no correlations went above 0.85 or below -0.85. With the target in this case being the opening weekend revenue, we can look directly at the correlations concerning this factor. In combination with the analysis of the factors visually, the correlation matrices will be useful for holistically analyzing each factor about revenue and opening weekend revenue.

4 Preliminary Analysis of Various Factors

The overarching goal of this project is to predict box office success. From a monetary standpoint, success depends on the revenue that a movie brings in. Thus we want to analyze what factors tend to bring in higher revenues, and if applicable, which of these factors lead to a movie producing a higher revenue with a relatively lower budget. Both the *Kaggle* data set and the Math512 set have many factors that can provide meaningful insight into this discussion. For these data sets, we utilized a general procedure. That being:

1. Determining the average revenues for each unique item in a factor
2. Graphing this information and analyzing any visual trends
3. Repeating steps 1 & 2 for the average Rev/Bud to assess how the revenue-budget ratio affects the plots

Due to the large sizes of unique actors, directors, and production companies from the *Kaggle* data set, the top 20 from each will be displayed and discussed here. The entire range of graphs will be provided in the second section of the appendix.

4.1 Genre

One important feature we analyzed from both the *Kaggle* and Math512 data sets was genre. More specifically, the primary, secondary, and third genre features from the *Kaggle* set and the genre category from the Math512 data set. Figures 6 and 7 display the information outlined in the procedure for the *Kaggle* set. Similarly, figures 8 and 9 display this information for the Math512 set. Note, however, that only 18 unique genres were listed in the *Kaggle* set and 11 unique genres in the Math512 data set.

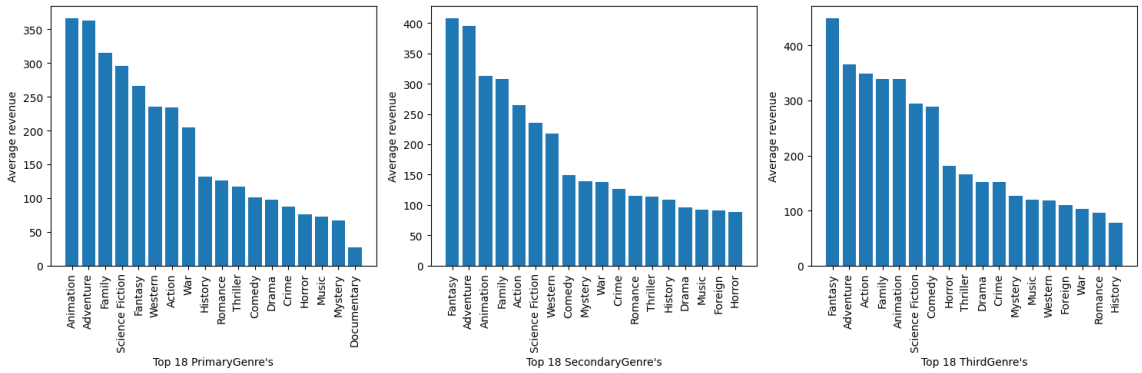

Figure 6: Plot of Average Revenue (millions) vs Genre from *Kaggle* set

Looking generally at all four of these figures, it is clear that some genres tend to outperform others for the average revenue and Rev/Bud features. In the case of Figure 6, the subplots of the primary, secondary, and third genres indicate that genres like animation, adventure, and fantasy have the highest average revenues.

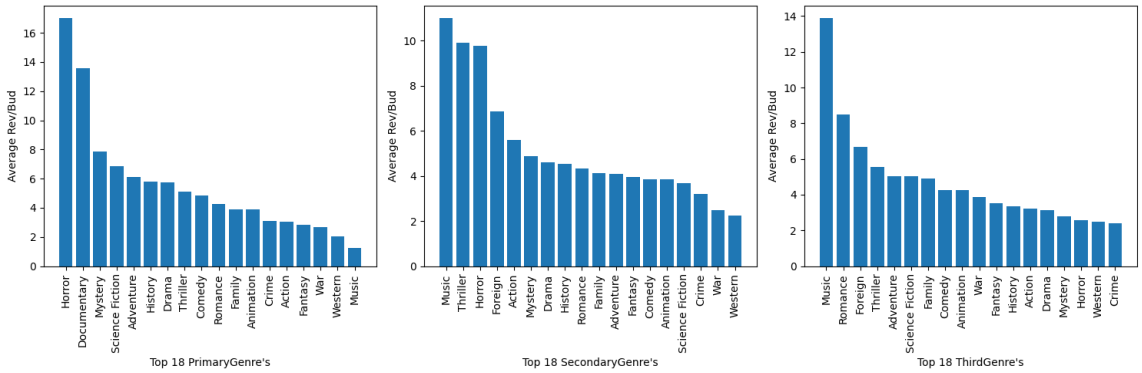

Figure 7: Plot of Average Rev/Bud vs Genre from *Kaggle* set

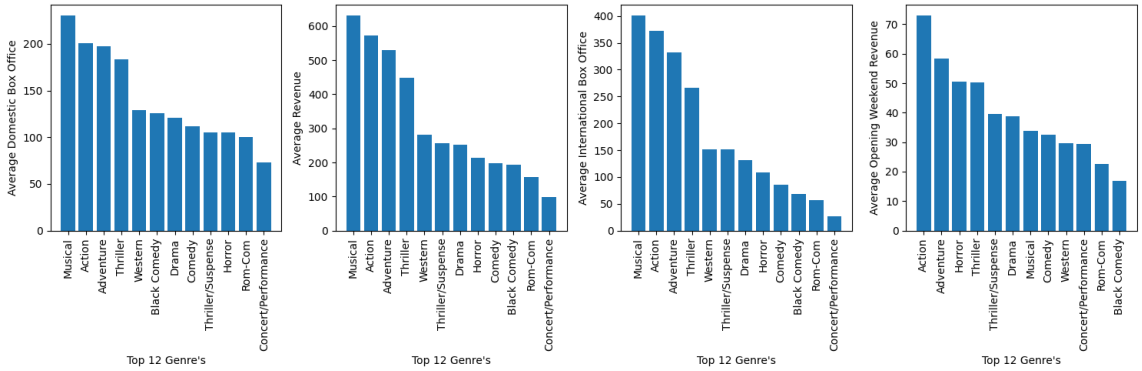

Figure 8: Plot of Average Revenues (millions) vs Genre from Math512 set

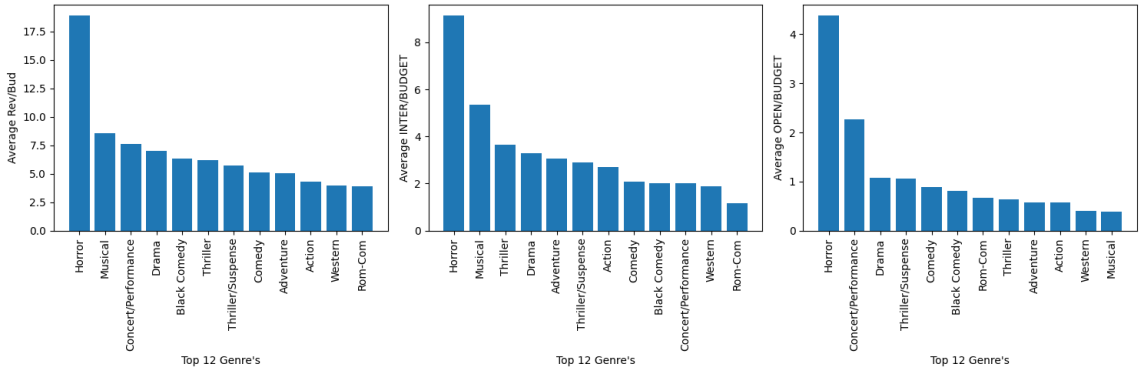

Figure 9: Plot of Revenue Ratios vs Genre from Math512 set

Particularly, fantasy holds the highest average revenue for both the secondary and third-genre features. In the case of Figure 8 for the Math512 data set, the genres of music, action, and adventure have the highest average revenue. Interestingly, musical holds the highest average in the total revenue, domestic, and international cases while being much lower for the opening weekend selection. This might indicate that the music genre, despite performing poorly during the opening weekend, has longevity in the theaters as the total revenue in the end exceeds all other genres. Comparing the results from these two data sets suggests at the very least that genres like adventure are top performers, while musical and fantasy hold even higher average revenues in the *Kaggle* and Math512 sets respectively.

Looking at Figure 7 from the *Kaggle* set, horror holds the highest average Rev/Bud ratio, being over 16 in the primary genre case. In the subsequent secondary and third genre selections, music and thriller as well as music and romance take the top spots. For the Math512 set, the Rev/Bud ratio averages are provided in Figure 9. In this figure, horror also holds the highest average Rev/Bud ratio, being nearly 18. It also has the highest Inter/Budget and Open/Budget ratios as well. Interestingly, musical and musically derivative genres like Concert/Performance take the second highest revenue ratio spots across all three subplots. Comparing the results of these ratio graphs across both data sets indicates that horror definitively has the highest Rev/Bud ratio out of all genres. However, music and musical type genres also tend to have large Rev/Bud ratios. This suggests at a minimum that, despite possibly smaller budgets for horror or musically categorized movies, they can make a profit many times the initial input. The analysis from all four of these figures also seems to oppose the notion that action is the highest-performing genre. If we are to take on the role of movie consultants, an action theme might not necessarily be the first movie genre we would attempt to pitch or approve. However, it still consistently is present in the top 7 average revenues across both data sets.

4.2 Actor

The second main feature we analyzed stemmed from the *Kaggle* set only, being the actor choice. More specifically we studied the lead, secondary, and third actor features. Figures 10 and 11 display the visualizations outlined in our procedure for the actor feature.

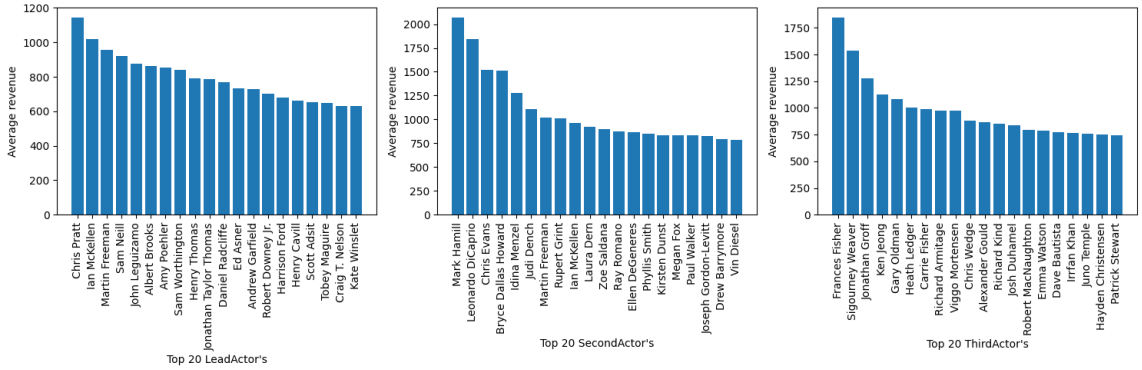

Figure 10: Plot of Average Revenue (millions) vs Actor from *Kaggle* set

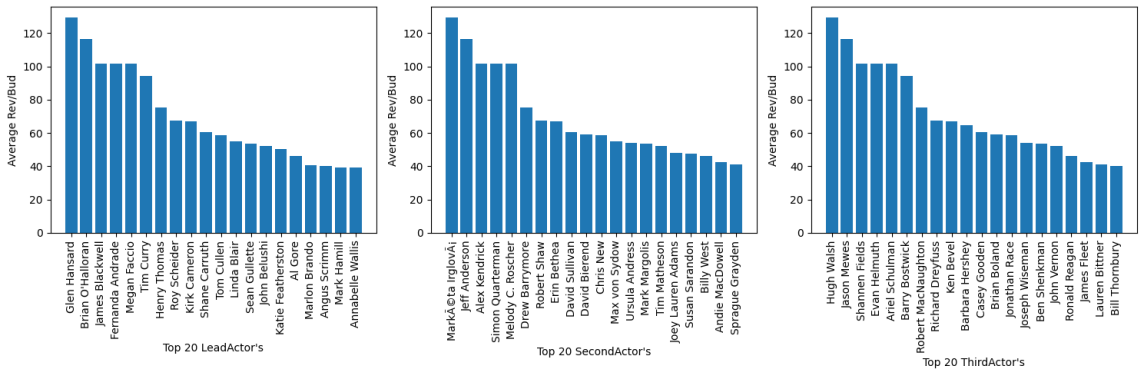

Figure 11: Plot of Average Rev/Bud vs Actor from *Kaggle* set

The size for this factor set was much larger, thus it is easiest to consider looking at the top 20 highest-performing actors in the average revenue and Rev/Bud selections. Contained within Figure 10 are the top

20 highest average revenue-earning actors for the lead, secondary, and third categories. The *Kaggle* data set indicates that Chris Pratt has the highest average revenue in the lead actor feature. Similarly, Mark Hamill and France Fisher hold the highest average revenues in the secondary and third actor categories respectively. In these subplots, as one might have expected, there are a lot of recognizable actors. Typically, these actors were a part of notable film franchises as well. Mark Hamill, for instance, famously portrayed Luke Skywalker in the *Star Wars* films. Chris Pratt was the lead for the highly successful *Jurassic World* franchise, lifted from the already culturally acclaimed *Jurassic Park* films. In general, the average graphs for the lead, second, and third actors provided in the appendix indicate that some actors just have higher average revenues. If we were taking on the role of a movie consultant, we could use our key when provided an actor, and either separate or identify on the graph how well this individual has performed in generating revenue compared to their peers. We could also push for a different selection if necessary. For instance, pulling separately from the top 20 selection, we decided to compare action stars Harrison Ford and Tom Hardy when selected as the lead actor. We found that Harrison Ford's average revenue was nearly $292 million, while Tom Hardy's was roughly $149 million. On paper, based on revenue history, Harrison Ford is the more profitable choice. However, the actor selection might also be budget-dependent.

We can look at Figure 11 which displays the top 20 average revenue-to-budget ratios for the lead, secondary, and third actor selections. The majority of the top performers in this case are not as well known. However, this can still be insightful in the sense that, given a low budget, the choice of an actor from this Rev/Bud graph could still lead to large profits. The history of these individuals from the *Kaggle* set might suggest untapped potential as their performances are likely to have contributed to the large Rev/Bud profits.

Overall in consideration of a choice of actor, the past revenue and rev/bud averages can be very useful in predicting how well a movie might perform. Thus it is clear that the choice of actor matters as some generate revenue much better than others.

4.3 Production Company

The third feature we peered into was that of the production company provided by the *Kaggle* set. More specifically we analyzed, the primary, secondary, and third production company features that we isolated. The results are displayed in Figures 12 and 13.

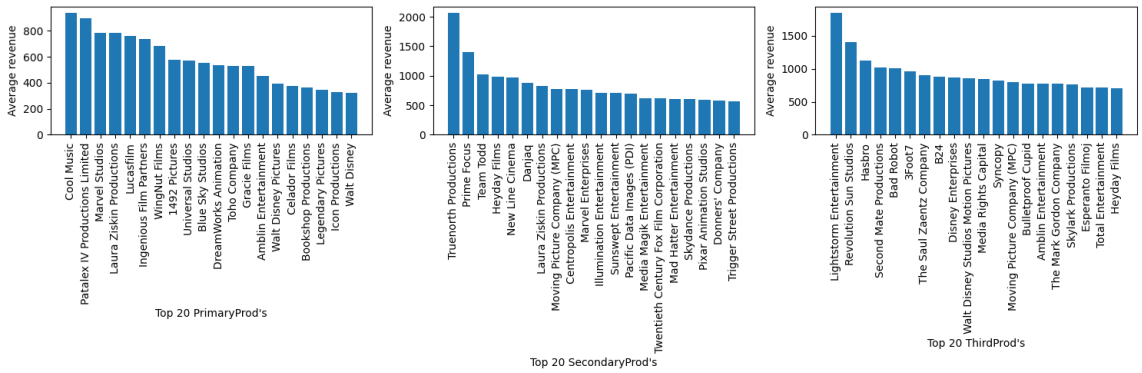

Figure 12: Plot of Average Revenue (millions) vs Production Company from *Kaggle* set

Like the actor selection, the size of the unique production companies was lengthy. Thus, we applied the same thought process by looking at the top 20 production companies concerning average revenue and average Rev/Bud. Similar as seen with the actor analysis, the average revenue vs production companies in Figure 12 display some recognizable production companies in addition to some lesser-known ones. For example, in the top 20 primary production companies, WingNut films (known for the Lord of the Rings Trilogy) and Marvel Studios are top contenders. Other notable companies such as Legendary Pictures and Universal Studios are present as well. However, there are also some unknown companies listed too. Cool Music Productions, for example, holds the highest average revenue for the primary production company according to the *Kaggle*

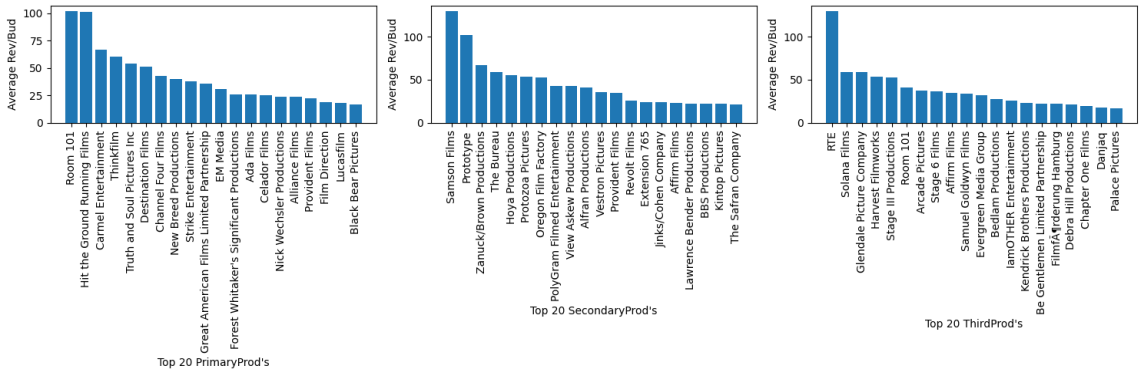

Figure 13: Plot of Average Rev/Bud vs Production Company from *Kaggle* set

data set. Researching this group indicates they specialize in music composition for various highly successful films. Thus as movie executive consultants, a recommendation of combining the Cool Music company for the music composition and another high-revenue company might lead to higher overall average revenue output.

Also similar to the actor feature analysis, the Rev/Bud subplots of Figure 13 display many lesser-known companies. This might indicate that on a low budget, these companies can still generate a large amount of revenue.

In general, the graphs from the appendix indicate that some production companies simply have averaged higher than others, and depending on whether or not this production company is the first, second, or third backing a movie, this also has consequences on the averages.

4.4 Director

The fourth feature we looked at was the director choice, again coming from the *Kaggle* data set. Following our procedure allowed us to generate the following figure:

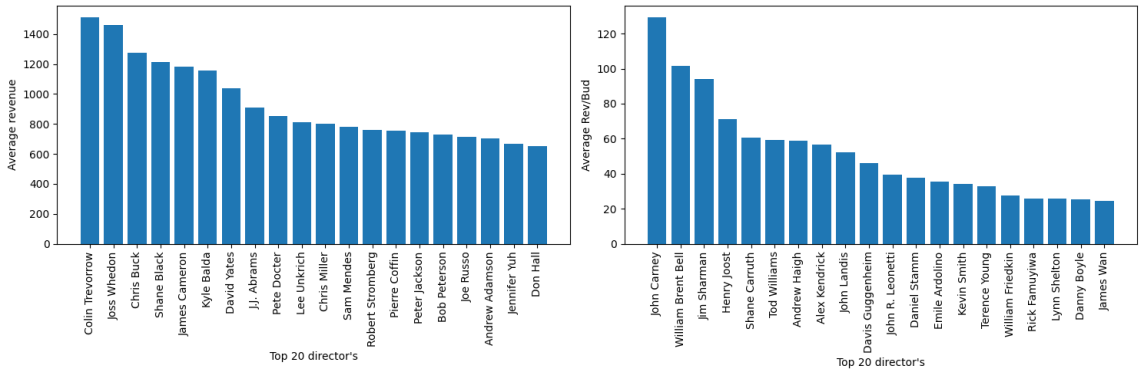

Figure 14: Plot of Average Revenue (millions) & Average Rev/Bud vs Director from *Kaggle* set

Just as with the actors and production company selections, the number of unique directors was also very long. Figure 14 indicates the top 20 directors concerning the average revenue and average Rev/Bud. Colin Trevorrow is displayed as the director with the highest average revenue while John Carney is indicated to be the director with the highest average Rev/Bud ratio. The same pattern continued from the actor and production company analysis, with the listing of the top 20 average revenues containing more well-known directors, while the top 20 average Rev/Bud contains lesser-known individuals. In estimating the revenue for a movie, we can consider the choice of director and use the average of the director compared to others to assess how well a movie will perform. Consider for example notable directors Robert Zemeckis and M.

Night Shyamalan who were not within the top 20 in either subplot of figure 14. Pulling from the total graphs provided in the index and using our "IntegerKey" list of directors, we can determine that Robert Zemeckis has an average revenue of $252 million. M. Night Shyamalan on the other hand has an average revenue of $193 million. While dependent on a variety of external circumstances, as a movie consultant, the revenue averages suggest choosing Zemeckis over Shyamalan for a larger profit. However, if a potential director is in our top 20 graphs, this could be further motivation to select that individual.

4.5 Runtime

Breaking away slightly from the procedure used for the previous factors, we looked at the run time from the *Kaggle* set. For the run time and revenue as well as the run time and Rev/Bud, the scatter plot took the following form in Figure 15:

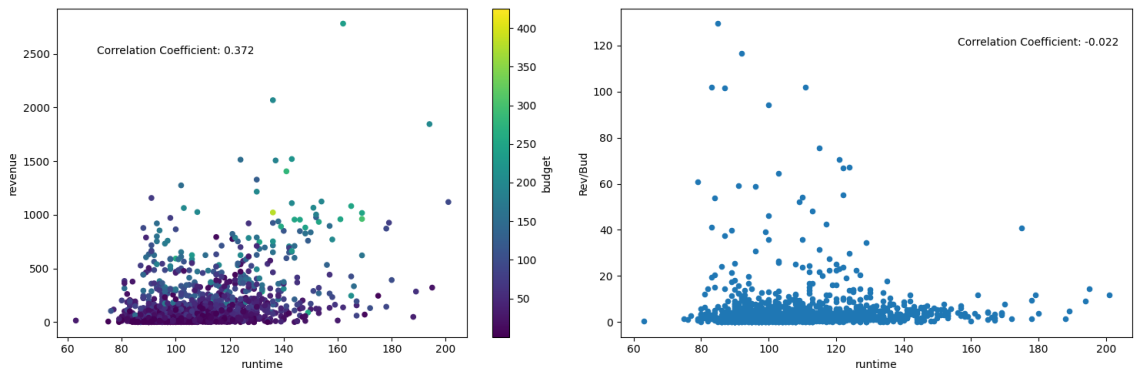

Figure 15: Scatter Plot of Revenue (millions) & Rev/Bud vs Runtime from *Kaggle* set

The points in both subplots are bunched together, not having a discernible pattern. As indicated in Figure 15, the correlation between the run time and revenue as well as the run time and Rev/Bud is weak. This is because the correlations are not exceeding either 0.5 or -0.5. Ideally, we would like to see correlations that are even larger than that which would indicate a strong relationship whether positive or negatively correlated. Overall this analysis from the *Kaggle* data set suggests that the run time does not seem to be an important factor in consideration of revenue and Rev/Bud.

4.6 Popularity Level

For the popularity level provided from the *Kaggle* data set, the procedural plots took the forms presented in Figure 16.

When analyzing the averages further for each case, there was a clear pattern, as seen in 1:

Table 1: Averages of Revenues and Revenue-to-Budget Ratios for Each Popularity Level

Popularity Level	**Average Revenue (Millions)**	**Average Rev/Bud Ratio**
Level 3	418	6.4
Level 2	160	5.9
Level 1	85	3.8
Level 0	42	3.9

Looking at Figure 16 and Table 1, it is clear that the higher the popularity level, the higher both the Rev/Bud and Revenue totals are. So based on this analysis, if the popularity level of a movie is higher, it will likely have a larger average revenue and larger average revenue-to-budget ratio.

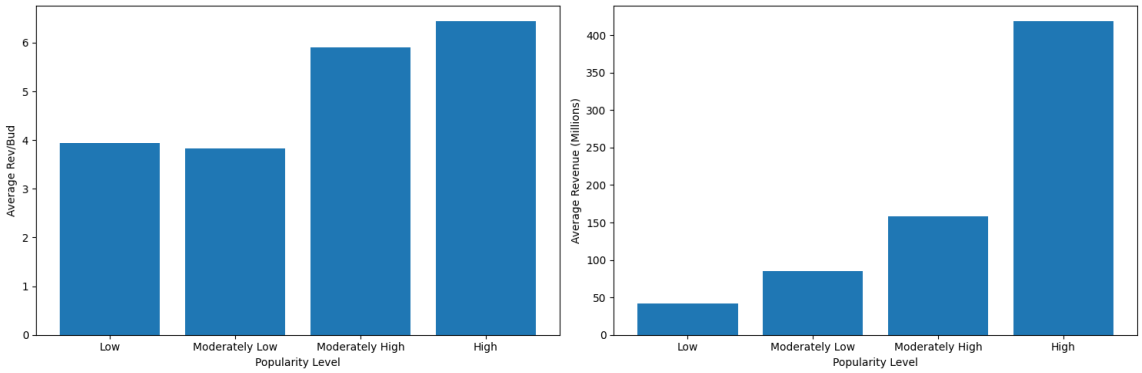

Figure 16: Plot of Average Revenue (millions) & Rev/Bud vs Popularity Level from *Kaggle* set

4.7 Google Trends Score

Another feature we analyzed, now from the Math512 set, was the Google Trends (GT) score. Figure 17 displays the scatter plots of the GT score feature and the corresponding revenues, now including opening weekend revenue. Similarly, figure 18 displays the scatter plot of the GT score and the revenue ratio values such as Rev/Bud.

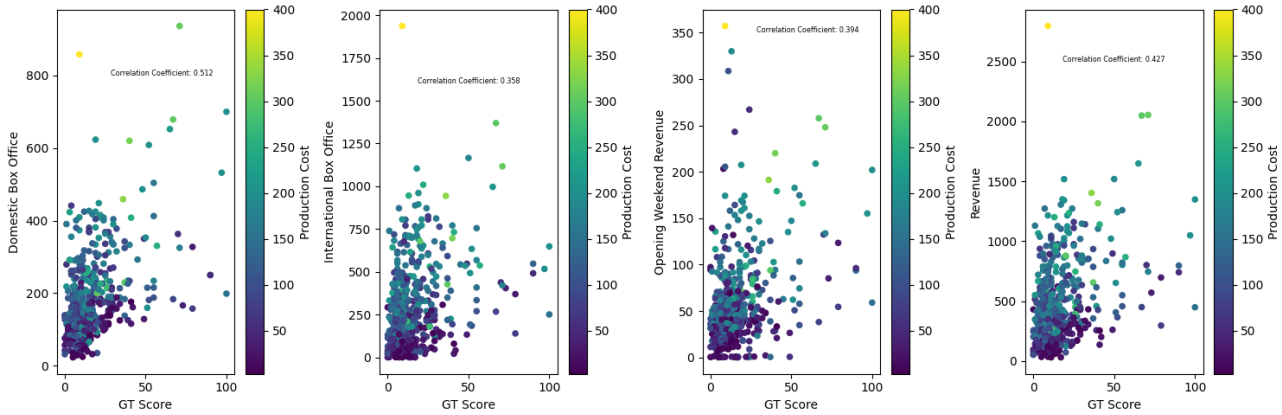

Figure 17: Scatter Plot of Revenues (millions) vs GT Score from Math512 set

Looking at the graphs of average revenue and average opening weekend revenue about the GT score, some notable trends emerge. The correlation between the GT score and domestic box office revenue is around 0.512. This indicates that there is some positive relation between the GT score and domestic revenues. The remaining revenues, however, have correlation coefficients that are smaller and do not exceed a 0.5 threshold. For instance, while the coefficient is about 0.5 for the domestic revenue, it is only 0.427 for the total revenue and 0.394 for the opening weekend. Even lower values are present in the scatter plots of Figure 18. This suggests a very low relationship between the GT score and the revenue ratios. Regardless, it would be best not to put too much faith into GT scores for predicting revenue beyond possibly the domestic revenue predictions.

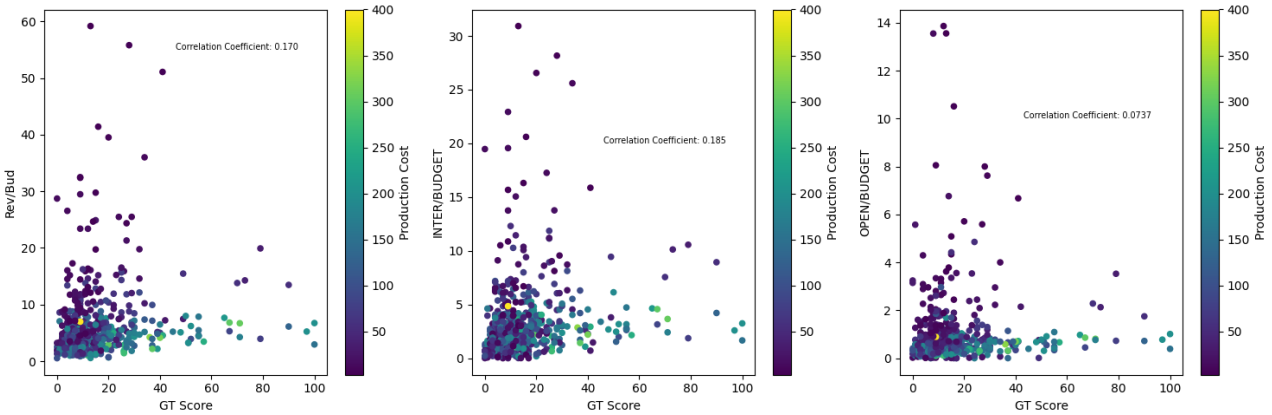

Figure 18: Scatter Plot of Revenues ratios vs GT Score from Math512 set

4.8 Rotten Tomato Ratings

The final feature we analyzed, again from the Math512 set, was the Rotten Tomato (RT) ratings. Figure 19 displays the RT rating following our procedural outline, also including opening weekend revenue. Similarly, figure 20 displays the RT rating with some additional revenue ratios beyond the Rev/Bud feature.

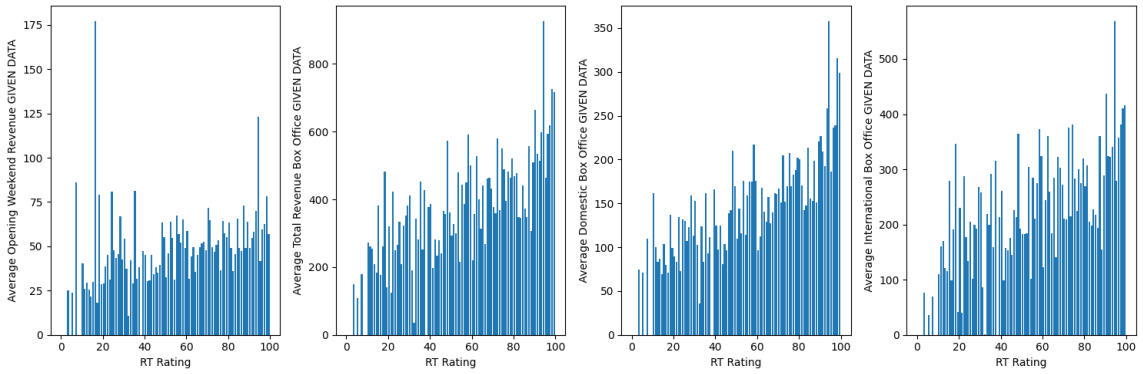

Figure 19: Plot of Average Revenues (millions) vs RT Rating from Math 512 set

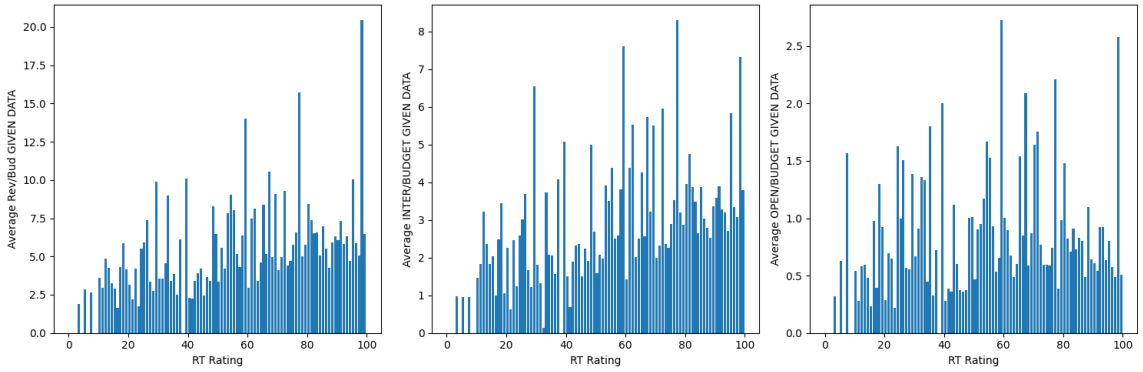

Figure 20: Plot of Average Revenue Ratios vs RT Rating from Math 512 set

Looking at the graphs of the RT ratings and average revenues in Figure 19, there are very distinct patterns that emerge. For the average revenue vs RT rating, the graph displays a left skew, where the higher the RT Rating is (which was multiplied by 100 to make it a percentage), the revenue also tended to increase. The average opening weekend revenue vs RT rating, however, tended to be flatter. This could be reflective of the fact that RT scores take time to be generated by the public, where for an opening weekend, possibly not all the reviewers are present to give a rating that matters. Either way, the average revenue vs RT rating does suggest that if the RT score is higher, the revenue will be too. The same trends were present in Figure 20 regarding the RT ratings and the revenue ratios such as Rev/Bud. The left skew again indicates the higher the RT rating is, the higher the Rev/Bud, Inter/Budget, and Open/Budget ratios are going to be.

5 Working with Neural Networks

One of the algorithms we looked at is neural networks, a machine-learning process that uses interconnected nodes or neurons in a layered structure that resembles the human brain. These are particularly useful in financial forecasting, marketing, and customer analytics. A neural network system is made up of several artificial neurons, or nodes, that take one or more inputs, for which they apply a mathematical operation and produce an output. In the context of our research, the input layer of the network is going to be made up of the features of movies that are most relevant to box-office outcomes. The hidden layer will utilize Feedforward neural network functions, also known as a multilayer perceptron (MLP), which can be used for regression tasks. Finally, the output layer will feature the predicted box office revenue. Our attempts at building an MLP neural network involved the use of *Keras*, an open-source library that provides a Python interface for artificial neural networks. The standard model consists of an input layer with 64 neurons and ReLU activation. ReLU, or Rectified Linear Unit, is the activation function:

$$f(x) = max(0, x) \tag{4}$$

This is a piece-wise linear function that returns the input value, x if it is positive and returns zero for negative outcomes. This is particularly useful to account for the positive relationships between input features and box office revenue. For example, as the budget or marketing expenditure increases, it is expected that the box office revenue will also increase. ReLU is well-suited for such situations because it allows the model to represent positive linear relationships effectively. When an input feature has a positive impact on revenue, the ReLU activation retains its positive values and supports the learning of this relationship. Because the first try with the standard Keras model resulted in an excessively high MAE of $187 million, several experiments involving the change of the number of hidden layers and neural nodes were executed. In the first modification, the number of nodes in the second hidden layer was increased from 32 to 128. The ReLU activation function was retained in this layer, while the output layer continued to use a linear activation function for regression tasks. The MAE was 183,958,714, which was lower than the previous model, indicating a better performance. The second modification introduced an additional hidden layer with 64 nodes, extending the model's depth and complexity. The MAE was 179,535,477, indicating a more significant drop than just increasing the number of nodes in the hidden layer. In the third modification, the activation function in the second hidden layer was replaced with the Exponential Linear Unit (ELU). ELU is an alternative to ReLU that has negative values which allows it to push mean unit activations closer to zero while achieving lower computational complexity. Its MAE ended up being 184,101,787. Looking at Figure 17, we can see that the lowest MAE value occurred with the model with the highest number of hidden layers. We saw a slight improvement in the model's performance through increasing the number of nodes as well. Finally, switching from ReLU to ELU functions did not lead to a significant change in the model's performance. Moving from this, we focused on simplifying other parts of the model by reducing the features analyzed by Keras from 12 to 3 (Budget, Director, Popularity). Furthermore, following the results of our experiment, we executed several trials with different numbers of increasing hidden layers and neural nodes. The most effective model, which resulted in the lowest calculated MAE had 7 hidden layers starting from 512 nodes and halving every consecutive layer until it reached 8 nodes (as shown in 21. The learning rate of the model was also adjusted to 0.0001, which led to better convergence. All these changes resulted in the plot at 22, with an MAE of about %51 million, which is about one-third of the one found at the beginning of the research.

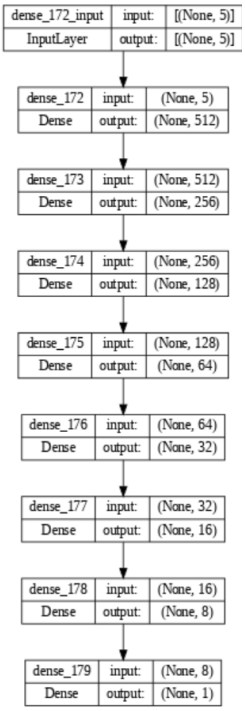

Figure 21: Neural Network Structure

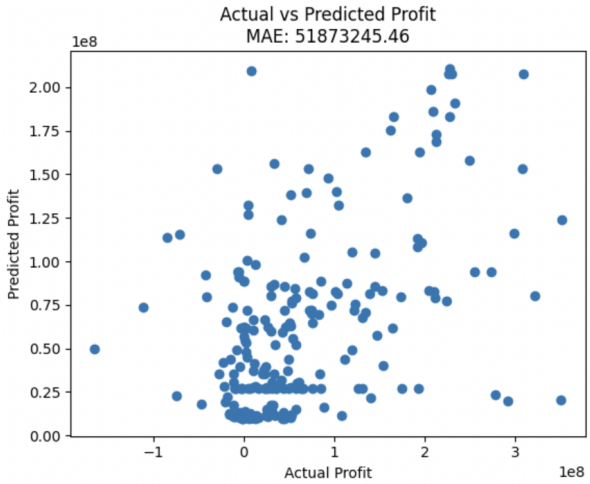

Figure 22: Plot of outcome for MLP Neural Network Prediction

In our research, we are also exploring a range of algorithms including k-NN, Multilinear Regression, and Miss Forest, each with unique strengths for predicting box office outcomes. The k-NN algorithm uses proximity to similar past data points to predict outcomes, leveraging patterns in movie success. Multilinear Regression examines how various features like budget and star cast influence box office revenue, using a linear equation model. Miss Forest, on the other hand, fills gaps in data, ensuring our analysis is comprehensive and reliable. This multi-algorithmic approach, combined with our MLP neural network, offers a holistic view of the movie industry, enhancing our predictive accuracy through diverse analytical methods and continuous refinement.

6 Applications of Machine Learning Algorithms

For the two datasets, we are going to use the machine learning regression algorithms of MissForest Regression and Multi-Linear Regression to try to create the best predictive model possible in projecting potential gross revenue, Domestic Box Office Revenue, International Box Office Revenue, and Opening Weekend Revenue.

6.1 Machine Learning on the Kaggle Dataset

Starting with the Kaggle dataset, the only feature that we are going to be attempting to predict is total revenue. To start we are going to take a look at the correlation matrix to try to eliminate any features with multicollinearity. Once we do so we utilize the MissForest package as well as multi-linear regression and we get the MF(MissForest) regression score of 0.8522 and a multi-linear regression score of 0.6806.

6.2 Machine Learning on the Math512 Dataset

For the Math512 dataset, we are first going to take a look at projecting total gross revenue.

We eliminate Domestic Box Office and International Box Office as they have multicollinearity factors greater than 0.85 or below -0.85. Upon doing so we utilize MF Regression and MLR to compute the accuracy scores of our current models with the data described previously throughout this paper. Our MF Model easily gives us the better score of the two models with a score of 0.9467, while our MLR model gives us a score of 0.7339.

Next, we try to create a model that best predicts total Domestic Box Office Revenue. We eliminate the respective features with multicollinearity and run our predictive models. The models give us an MF score of 0.8978 and an MLR model score of 0.8186.

We see that once more our MF predictive model is better, but this time the two models are much closer together.

Next, we create a model for estimating total International Box Office revenue. After running preprocessing steps we receive an MF score of 0.8564 and a MLR score of 0.7221.

As we can see once more the MF model is far superior to the MLR model although each of the scores has decreased from the strong results of the prior model.

The last revenue stream we create a model for is Opening Weekend Revenue. We run our preprocessing steps and feature elimination before receiving an MF score of 0.8787 and an MLR score of 0.5192.

6.3 Thoughts on the Models

After looking at each of the models, MissForest was stronger in every single case. Although it was not able to approach the 0.90 threshold that we had hoped for in all instances, it was very close with the lowest accuracy being roughly 0.85.

7 Summer Success?

One of the main tasks we were asked to address is to support or refute the notion that summer months are the best time to open a movie as that is when many people visit theaters. To do so, we have to define what makes a time frame optimal. Specifically, we will look at the totaled revenues and opening weekend revenues across the whole year. If there is a trend for higher revenue values during the summer months, this could be indicative that the statement is true. Simply put, a time frame is optimal if revenues are higher during this period compared to the rest of the year.

To address this question, we are going to utilize the CR values and corresponding revenues from the first two data sets. Since CR values represent every day in the given year, it becomes much easier to visualize trends using a week-by-week basis instead. This is accomplished through a simple procedure:

1. Group CR values in weekly intervals (52 weeks total)
2. Keep track of the revenues corresponding to each CR

3. Sum or average the revenues within a given week

4. Graph this across 52 weeks

We enacted this procedure using our created function "split52()" which takes in a data set, a feature column we want to group its CR values by 52 weeks, and then the option to sum or add the feature column values. It produces an array with either the sum or averages for a week placed at the proper index. I.e index 0 would correspond to the average or sum of the 1st week. The true week number can be shifted later by 1 graphically.

Implementing this procedure for the *Kaggle* data set with regards to revenue produced the following figure:

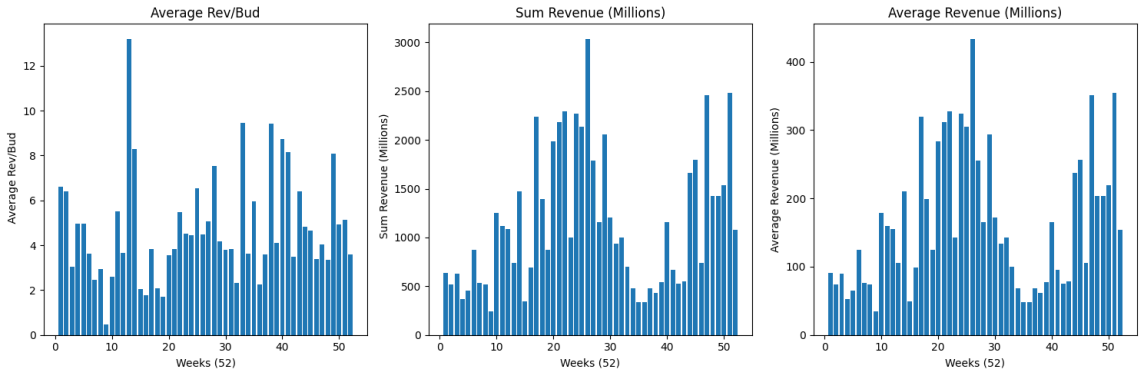

Figure 23: Average & Sum Revenues (millions) & Average Rev/Bud across 52 Weeks for the *Kaggle* data set

Implementing this procedure for the Math512 data set offered more graphical options, despite a smaller data set size to work with:

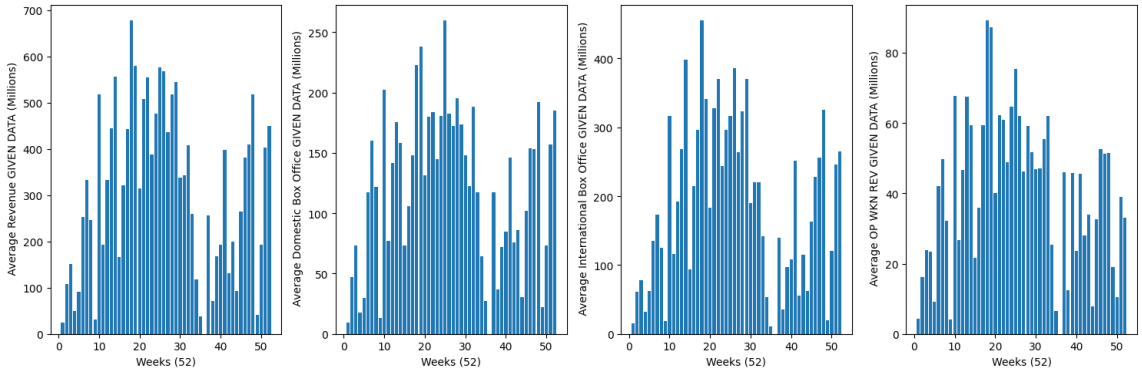

Figure 24: Average Revenues (millions) across 52 Weeks for Math512 data set

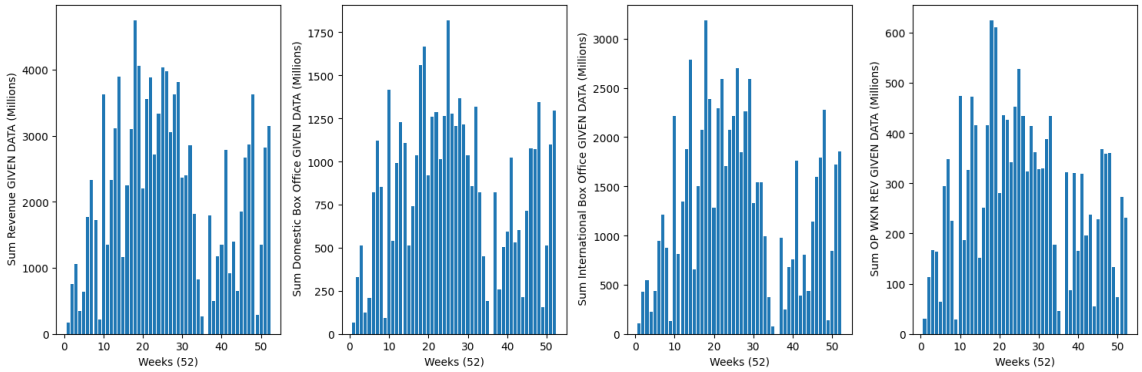

Figure 25: Sum Revenues (millions) across 52 Weeks for Math512 data set

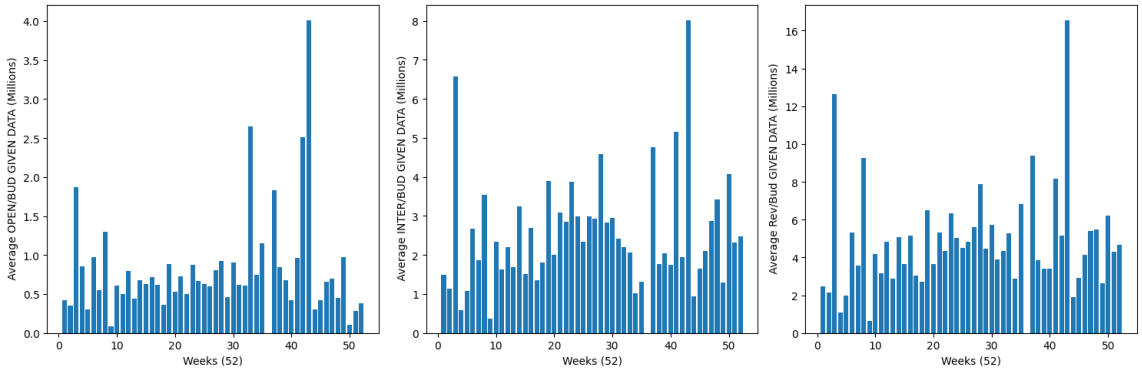

Figure 26: Revenues Ratios across 52 Weeks for Math 512 data set

Looking at Figure 23 corresponding to the *Kaggle* data set, in both the average and summed revenues across the 52 weeks, there is a net spiked area between weeks 18 and 33. There is also a distinct drop in revenues quickly after the 33-week mark and before week 16. Similarly, when looking at the sum and average revenues from the Math512 data set in figures 24 and 25, there is also a distinct drop around these same weeks. The time frame between them also displays much higher revenue values. Another feature of these graphs is that the revenues past week 40 tend to rise again until the end of the year. Compare that to the much lower revenues displayed in the first few weeks of the year.

The opening weekend revenue subplots show a similar shape, albeit not as apparent. Within the 18 to 33 inclusive week range, the opening weekend revenues are flatter, with a notable spike at the 18-19 week range. It too shows smaller revenues within the first few weeks, a stark contrast to the revenues in the middle of the year.

In general, the weeks that are considered a part of the summer months vary based on culture. However, if we define the summer period from a commercial perspective using the unofficial start and end of summer (Memorial Day to Labor Day respectively), this corresponds to weeks 21 through week 35. With this in mind, the graphs display these larger spikes in revenue within the time frame. This is especially apparent in the graphs from the *Kaggle* set. For the Math512 data, this also appears to be the case. Possibly, with more entries like the first set, the Math512 data might take on the *Kaggle* data set shape more distinctly. Either way, the higher trend in revenue within this time frame seems to support the notion that summer months are the best time to open a movie. However, the higher revenue trend within 5 weeks of the 52-week mark seems to also indicate that the holiday season might also be an optimum time for movies, though the revenue peaks are not nearly as large as those in the summer months time range across all graphs.

8 The Impact of China on International Box Office Revenue

To determine whether or not success at the Chinese box office is required to be successful at the international box office, we must attempt to quantify or understand what it means for a movie to do well internationally. Specifically, we are going to look at the average total international revenue of movies when China is and isn't the top international market. However, there are many things that we need to do before we can get to this step.

To answer this question we are going to be using *the-numbers* data set that we collected as specified earlier in the paper. To ensure truly international films were being siphoned for our subsequent analysis, we applied a filter to eliminate instances in the data set in which the "Share of Number One Market" column value was greater than 51. Since the scraper pulled movies that were shown internationally, this also included films that were domestic to another country other than the United States. Once filtered, the data set totaled 1073 movies released between the years of 2013-2023. We can then look at the data set and the number of times each country is a top market as well as its generally generated revenue. Doing so we can see that the countries that are most commonly the top international markets are the UK & China. This is also documented in the appendix.

First, we are going to create two different data frames, one that includes the rows in which China is the number one international market, and one that includes the rows in which China is not the number one international market. We do this through a simple process:

1. Set a condition i.e: Number One Market is or isn't China
2. Assign all rows that satisfy that condition to a new data frame

Once we do this we find that there are 212 movies where China is the top international market and 861 movies where it is not.

Now we look at three different statistics for each data frame.

1. Market Share average of the top market: International Revenue generated by the top Market / Total International Revenue
2. Box Office revenue of the top international Market (millions of dollars)
3. Average Total International Revenue (millions of dollars)

These statistics gave us these results:

Top International Market	**China**	**Any other country**
Top Market Average Market Share	25.44%	17.73%
Box Office Revenue of Top International Market (in millions)	$69.65	$13.42
Average Total International Revenue (in millions)	$271.21	$97.04

We can immediately see huge differences between the two data sets. There is a roughly 133% increase in the top international market share of the top international market when it is China going from roughly 20% to 30%. We can also see that the box office revenue of the top market when it is China is roughly $69 million as opposed to $13 million when it isn't in China, almost a 600% increase. Lastly, we can see a roughly $174 million difference when China is the top international market for total international box office revenue, a roughly 279% increase in International revenue generated.

Compare that to a similar analysis done with the UK seeing as it is most commonly the top international market. The statistics give us about 20% for the UK market share average when it's the top international market and roughly 19 million when the UK's average box office revenue is the top international market. These results show us the enormous difference that occurs when a movie's top market is China as opposed to the UK.

From these results, we can argue that China is a very important factor in movies performing well in the international market. Although movies may not necessarily require significant success at the Chinese box office to succeed internationally, it can certainly be a significant boost in total International Box Office

Revenue if the market is properly reached. With this in mind, we argue that studio executives should do all that they can to market their movies well to Chinese audiences.

9 The Importance of a Strong Opening Weekend

It is very easily and naturally understood that a strong opening weekend is vital for the success of a movie. We can say this via our intuition about the importance of a good early reception of the movie from audiences. But just how important is an Opening Weekend in determining the success of a film? For this, we are going to take a look at the data from our Math512 dataset.

In the Math512 dataset, we are going to take a look at our Rev/Bud ratios as a way to group movies together, marking 5 different barometers of movies based on their Rev/Bud score. These barometers are below 1.5, below 3, below 5, below 7, and greater than 7. These give us the resulting counts in the table below. We then divided the Opening Weekend Revenue by the total revenue generated to determine its significance. The count, median, and average of each are displayed in the table below.

Rev/Bud Score	**Count**	**Median**	**Average**
Over 7	141	0.1069	0.1133
Between 5 & 7	94	0.1070	0.1071
Between 3 & 5	193	0.1117	0.1285
Between 1.5 & 3	161	0.1129	0.1337
Under 1.5	30	0.1629	0.2619

Interestingly enough it seems that as the movie becomes more successful, the impact of its opening weekend seems to matter less as movies are more and more successful, suggesting that movie success is more tied to success in the long run than that of immediate success.

10 Conclusion

The Box Office has always been a phenomenon in the generation of capital, and trying to predict it is a task that many groups have set out to do. Our group has investigated an extensive collection of movies in an attempt to predict their success and we reached several conclusions in our investigations.

When looking at specific features, we found that the top genres in terms of total average revenue were Animation, Adventure, Family, Musical, and Action among others. When specifically looking at the "best bang for your buck" or how much a movie can return with minimal investment in budget, we found that Horror movies performed exceptionally well. Documentary and musical films also followed this trend. Movie genres with the lowest return on budget included Romantic Comedies, movies on war, and the Western genre.

When trying to analyze the importance of actors to films' profit generation we found that specific actors such as Chris Pratt correlated to significant revenue growth. Furthermore, actors France Fisher and Mark Hamil also contributed greatly to the revenues generated by movies when not listed as the primary protagonists, thus displaying how their role as secondary characters could contribute to a movie's success.

Production companies were another feature that we put under the microscope. We found that using specific production companies such as WingNut Films (renowned for their production of the Lord of the Rings Trilogy) as well as Marvel Studios saw the production of very high-quality movies with equally high revenue generated. In total, we were able to ascertain a plethora of additional movie studios that could be employed to maximize revenue generation beyond the top 20.

Directors were a feature very similar to production companies and actors. We were able to map the top movie directors and have identified directors such as Joss Whedon and Colin Trevorrow as the two most prolific directors, among others. Run-time was something different altogether. We found that it did not have any particularly discernible patterns although we did find several outliers within the 80-130 minute time frame. This could suggest that movies straying just around an hour and a half to two hours could generate more revenue.

For popularity level, our findings align with what we initially expected: movies with higher popularity had significantly higher revenue. We also saw that movies with a moderate amount of popularity could generate similar rates of return on budget, thus suggesting that movies with the highest popularity spent a significant amount of money on their budget which ate into their profits. We found that Google Trends score had a small positive correlation with predicting revenue generation, suggesting that more traffic on Google regarding a movie would indicate more revenue generated. The relationship was not as strong as we expected it to be. Similar to popularity level, we also looked at Rotten Tomato's ratings and found that movies with higher Rotten Tomato scores tended to generate more money. Features like RT ratings and popularity levels highlight the importance of positive movie reception in the generation of total long-term revenue.

With the use of Machine Learning, we found that the Miss Forest prediction models allowed us to generate the best model scores. We attained revenue prediction success of 85.22% on our *Kaggle* data set, a 94.67% when trying to predict total gross revenue on the Math512 data set, as well as 89.78% accuracy of Domestic revenue projections, 85.64% accuracy of International revenue projections, and 87.87% accuracy of Opening Weekend revenue projections.

When determining the importance of a movie's release date we found that movies released in the summer months tend to make more money when compared to other months. The second best time to release movies appeared around the holiday season at the end of the year, which was followed by a large revenue slump in January. Given this secondary trend, we can conclude that studio executives should specifically avoid releasing movies in January as there is a history of poor box office performance within this month. We also found that China is critical in a movie's box office success as it has the potential to significantly propel International Gross Revenue. Movies that saw China as its biggest international market saw a $174 million increase in international gross revenue when compared to movies that had bigger international markets elsewhere. This was a 279% increase in the international box office.

The last thing we looked at was the importance of a strong opening weekend, and we found that a movie's opening weekend did not seem to impact a movie's success too severely.

To draw our conclusions we had to make a few assumptions. For example, inflation was not accounted for in our calculations of the Math512 and *the-numbers* data sets because of the extensive data pre-processing effort and time it would have entailed. The oldest movies in these data sets go back to the year 2000, where $1 today would have accounted for $1.75. We recognize that this flaw put older movies in our catalog at a disadvantage for a large part of the analysis, therefore we tried to focus more on the budget/revenue ratios to even out the field. Furthermore, we assumed the *Kaggle* data set retrieved to be accurate when Kaggle is a platform that allows users to modify and publish any data they want.

Although we did our best to come to our conclusions and to investigate our results, we came across many problems while working with our data set that remain unresolved. One of these problems was to look at the spread of revenue throughout a movie's lifetime at the Box Office. We were unable to find adequate data to answer this question and instead just looked at how important a movie's opening weekend was. We also ran into another issue during our analysis of the Chinese box office impact. We ideally would have liked to include the U.S. market in the total international revenues as the two largest markets are both the U.S. and China for any movie. *the-numbers.com* that we collected the information from did not include this option, nor a more substantial breakdown of each market's share of the total international revenue. It only included the top market share, which was still valuable.

Had our group had even more time to analyze and work with our data we would have tried to make a multitude of improvements. Expanding section 9 is one immediate example. If our group had had more time we would have taken a deeper look at determining the changes of revenue from week to week rather than just from Opening Weekend to the total revenue generated. This investigation would have given us a much better look at the importance of immediate box office success/the importance of staying power for each movie. Furthermore, with more time our group could have explored a different library than Keras for Neural Networks, one more computationally efficient and that could support the analysis of more movie features. Another thing that we would have tried to do is increase our data size, particularly for our predictive models as well as our analysis of the impact of the Chinese Box Office and the importance of the release date. Although we were fairly happy with the size of our data, more data is never a bad thing so long as that data is accurate and can work cohesively with the information that we had already attained. The results of working with more data would have allowed us to potentially come to stronger conclusions, or enhance our

findings.

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11 Appendix

11.1 Integer Index Tables for All Data sets

Table 2: Kaggle Key

Master Index	Genre	Actor	Production Company	Director
0	Action	Chris Pratt	Universal Studios	Colin Trevorrow
1	Adventure	Tom Hardy	Village Roadshow Pictures	George Miller
2	Western	Shailene Woodley	Summit Entertainment	Robert Schwentke
3	Science Fiction	Harrison Ford	Lucasfilm	J.J. Abrams
4	Drama	Vin Diesel	Universal Pictures	James Wan
5	Family	Leonardo DiCaprio	Regency Enterprises	Alejandro GonzÃ¡lez IÃ±Ã¡rritu
6	Comedy	Arnold Schwarzenegger	Paramount Pictures	Alan Taylor
7	Crime	Matt Damon	Twentieth Century Fox Film Corporation	Ridley Scott
8	War	Sandra Bullock	Walt Disney Pictures	Kyle Balda
9	Fantasy	Amy Poehler	Columbia Pictures	Pete Docter
10	Thriller	Daniel Craig	DNA Films	Sam Mendes
11	Romance	Mila Kunis	Marvel Studios	Lana Wachowski
12	Mystery	Domhnall Gleeson	Double Feature Films	Alex Garland
13	Horror	Adam Sandler	Studio Babelsberg	Chris Columbus
14	History	Robert Downey Jr.	New Line Cinema	Joss Whedon
15	Animation	Samuel L. Jackson	Focus Features	Quentin Tarantino
16	Documentary	Liam Neeson	Participant Media	Olivier Megaton
17	Music	Paul Rudd	Gotham Group	Peyton Reed
18	blank	Jennifer Lawrence	Element Pictures	Francis Lawrence
19	Foreign	Britt Robertson	Wildgaze Films	Brad Bird
20		Dwayne Johnson	Lakeshore Entertainment	Brad Peyton
21		Dakota Johnson	Bazelevs Production	Sam Taylor-Johnson
22		Christian Bale	Haut et Court	Adam McKay
23		Tom Cruise	Killer Films	Christopher McQuarrie
24		Taron Egerton	Waverly Films	Matthew Vaughn
25		Mark Ruffalo	Endgame Entertainment	Tom McCarthy
26		Dylan O'Brien	Lionsgate	Wes Ball
27		Sharlto Copley	Parts and Labor	Neill Blomkamp
28		Jack Black	Kramer & Sigman Films	Rob Letterman
29		Brie Larson	Imagine Entertainment	Lenny Abrahamson
30		Raymond Ochoa	Working Title Films	Peter Sohn
31		Saoirse Ronan	Blue Sky Studios	John Crowley
32		O'Shea Jackson Jr.	Sony Pictures	F. Gary Gray
33		Blake Lively	Universum Film (UFA)	Breck Eisner
34		Jim Parsons	TriStar Pictures	Lee Toland Krieger

Kaggle Key 2 continued from previous page

Master Index	Genre	Actor	Production Company	Director
35		Colin Farrell	Temple Hill Entertainment	Ilya Naishuller
36		Cate Blanchett	Blumhouse Productions	Tim Johnson
37		Robert De Niro	Onyx Films	Yorgos Lanthimos
38		Jason Clarke	Amazon Studios	Todd Haynes
39		Edgar RamÃrez	Wonderland Sound and Vision	Nancy Meyers
40		Bradley Cooper	Alliance Films	Baltasar KormÃ¡kur
41		Ryan Reynolds	Denver and Delilah Productions	Ericson Core
42		Johnny Depp	Bold Films	John Wells
43		Chris Hemsworth	Forest Whitaker's Significant Productions	Tarsem Singh
44		Miles Teller	BBC Films	David Koepp
45		Anya Taylor-Joy	Little Stranger	Michael Mann
46		Melissa McCarthy	EuropaCorp	Josh Trank
47		Will Smith	Legendary Pictures	Robert Eggers
48		Rupert Friend	Ingenious Media	Paul Feig
49		Michael B. Jordan	Shangri-La Entertainment	Glenn Ficarra
50		Noah Schnapp	Red Crown Productions	Aleksander Bach
51		Joseph Gordon-Levitt	Provident Films	Ron Howard
52		Keanu Reeves	Centropolis Entertainment	Ryan Coogler
53		Steve Martin	Thunder Road Pictures	Brian Helgeland
54		Marie Avgeropoulos	WingNut Films	Steve Martino
55		Jason Bateman	Black Bear Pictures	Peter Landesman
56		Will Ferrell	Scott Free Productions	Jonathan Levine
57		Jeff Bridges	Toho Company	Eli Roth
58		Kate Winslet	Irish Dreamtime	Jessie Nelson
59		Mae Whitman	DreamWorks Animation	Robert Zemeckis
60		Shannyn Sossamon	Davis-Films	Daniel Benmayor
61		Jennifer Lopez	The Weinstein Company	Henry Hobson
62		Tye Sheridan	Echo Films	Joel Edgerton
63		Olivia DeJonge	Open Road Films	Sean Anders
64		Adam Scott	Stoney Lake Entertainment	Mark Osborne
65		Charlize Theron	Ghoulardi Film Company	Jocelyn Moorhouse
66		Owen Wilson	Miramax Films	Ari Sandel
67		Shameik Moore	Automatik Entertainment	Ciaran Foy
68		Helen Mirren	South Australian Film Corporation	Rob Cohen
69		Tina Fey	Hopscotch Features	Christopher B. Landon
70		Jonny Weston	Head Gear Films	M. Night Shyamalan
71		Ed Skrein	Warner Bros.	Michael Dougherty
72		Mia Wasikowska	WhiteFlame Productions	Gilles Paquet-Brenner
73		Chiwetel Ejiofor	DC Comics	John Erick Dowdle

Kaggle Key 2 continued from previous page

Master Index	Genre	Actor	Production Company	Director
74		Tom Kenny	Le Pacte	Rick Famuyiwa
75		Dermot Mulroney	Media House Capital	Simon Curtis
76		Maggie Smith	Alcon Entertainment	Jason Moore
77		James McAvoy	Worldview Entertainment	Dean Israelite
78		Julianne Moore	Pure Flix Entertainment	Camille Delamarre
79		Bill Murray	Eon Productions	Guillermo del Toro
80		Sally Field	Ingenious Film Partners	Billy Ray
81		Olivia Wilde	Fox Searchlight Pictures	Paul Tibbitt
82		Cassidy Gifford	Touchstone Pictures	Leigh Whannell
83		Aubrey Peeples	Walt Disney	Nicholas Hytner
84		Priscilla C. Shirer	Media Rights Capital	Paul McGuigan
85		Jeremy Irvine	Impact Pictures	Peter Sollett
86		Matthew McConaughey	Liberty Films UK	Barry Levinson
87		Chris Evans	Jerry Bruckheimer Films	Michael Showalter
88		Martin Freeman	2929 Productions	David Gelb
89		Scott Adsit	Dune Entertainment	Travis Cluff
90		Benedict Cumberbatch	Plum Pictures	Jon M. Chu
91		Ben Affleck	Smokewood Entertainment Group	Alex Kendrick
92		Hugh Jackman	DreamWorks SKG	Roland Emmerich
93		Scarlett Johansson	Section Eight	Christopher Nolan
94		Megan Fox	Paramount Vantage	James Gunn
95		Jake Gyllenhaal	Icon Productions	Joe Russo
96		Mark Wahlberg	GK Films	Chad Stahelski
97		Andrew Garfield	StudioCanal	Peter Jackson
98		Aaron Taylor-Johnson	The Mark Gordon Company	Don Hall
99		Angelina Jolie	Vidhu Vinod Chopra Productions	Morten Tyldum
100		Pierce Brosnan	Spyglass Entertainment	David Fincher
101		Jay Baruchel	Werc Werk Works	Neil Burger
102		Sylvester Stallone	Octagon Films	Bryan Singer
103		Will Arnett	HimenÃ³ptero	Luc Besson
104		Meryl Streep	Chris Brinker Productions	Jonathan Liebesman
105		Sullivan Stapleton	Nu Image Films	Dan Gilroy
106		Jonah Hill	Dimension Films	Michael Bay
107		Nicole Kidman	Fortress Features	Damien Chazelle
108		Jennifer Aniston	Gato Negro Films	Marc Webb
109		Jon Favreau	Smartest Man Productions	Doug Liman
110		Nicolas Cage	Isle of Man Film	Clint Eastwood
111		Oscar Isaac	Oscilloscope Laboratories	Gareth Edwards
112		Russell Crowe	Mark Johnson Productions	Robert Stromberg

Kaggle Key 2 continued from previous page

Master Index	Genre	Actor	Production Company	Director
113		Ryan Guzman	DreamWorks	Roger Donaldson
114		Joaquin Phoenix	American Zoetrope	Dean DeBlois
115		Joel Kinnaman	Devonshire Productions	Patrick Hughes
116		Mickey Rourke	Destination Films	Phil Lord
117		Jesse Eisenberg	Echo Lake Productions	Sergey Bodrov
118		Jack O'Connell	Screen Gems	Rob Marshall
119		Brenton Thwaites	Sahara One Motion Pictures	Noam Murro
120		Dane Cook	Magnolia Pictures	Phillip Noyce
121		Idris Elba	Capitol Films	Paul King
122		Perdita Weeks	Ghost House Pictures	Daniel Barnz
123		Annabelle Wallis	Great Scott Productions Pty. Ltd.	Jon Favreau
124		Aaron Eckhart	Millennium Films	Vic Armstrong
125		Essie Davis	The Zanuck Company	J.C. Chandor
126		Steve Carell	20th Century Fox	Darren Aronofsky
127		Jon Hamm	Bedford Falls Productions	David Dobkin
128		Kristen Bell	Polish Film Institute	Trish Sie
129		Evan Rachel Wood	Gaumont	Paul Thomas Anderson
130		Michelle Monaghan	Tree Line Films	JosÃ© Padilha
131		Christoph Waltz	Hunting Lane Films	Frank Miller
132		Samantha Morton	Escape Artists	Carlos Saldanha
133		Morgan Freeman	Mandate Pictures	George Clooney
134		Greg Kinnear	Matador Pictures	Angelina Jolie
135		Anne Hathaway	ApipoulaÃ¯	Theodore Melfi
136		Sarah Drew	Sony Pictures Classics	William Eubank
137		Kevin Sorbo	Katalyst Films	Roberts Gannaway
138		Mark Hamill	United Artists	Sam Miller
139		Roger Moore	Hammer Film Productions	John R. Leonetti
140		Sam Worthington	Castle Rock Entertainment	Stuart Beattie
141		Brad Pitt	Rainforest Films	Jennifer Kent
142		Daniel Radcliffe	Spring Creek Productions	Akiva Goldsman
143		Ed Asner	Vertigo Entertainment	Russell Crowe
144		John Cusack	Josephson Entertainment	Miguel Arteta
145		Tom Hanks	PathÃ© Pictures International	Jalmari Helander
146		Anika Noni Rose	Olympus Pictures	Craig Gillespie
147		Kristen Stewart	NALA Films	Rob Thomas
148		Dennis Quaid	Stone Productions	Andrew Fleming
149		Katherine Heigl	Bad Robot	Jay Oliva
150		Bill Hader	Oceana Media Finance	Terry Gilliam
151		George Clooney	Strike Entertainment	Steven Bernstein

Kaggle Key 2 continued from previous page

Master Index	Genre	Actor	Production Company	Director
152		Dakota Fanning	River Road Entertainment	Charles Martin Smith
153		Ben Stiller	North by Northwest Entertainment	Randall Wallace
154		Zac Efron	Studio 37	Kate Barker-Froyland
155		Gary Oldman	Mandalay Vision	Andrew Erwin
156		Rachel Weisz	Rogue Pictures	Rupert Wyatt
157		Kevin James	Anonymous Content	Harold Cronk
158		Cameron Diaz	Relativity Media	George Lucas
159		Jason Statham	Canal+	Lewis Gilbert
160		Ben Foster	Wildwood Enterprises	James Cameron
161		Alison Lohman	DRO Entertainment	David Yates
162		Sam Rockwell	Birdsong Pictures	Bob Peterson
163		Denzel Washington	Lions Gate Family Entertainment	McG
164		Isla Fisher	De Line Pictures	Guy Ritchie
165		Gerard Butler	This Is That Productions	Todd Phillips
166		Vince Vaughn	Maguire Entertainment	Ron Clements
167		Viggo Mortensen	Omnilab Media	John Lee Hancock
168		Seth Rogen	Microsoft	Ruben Fleischer
169		Bobby Campo	Serendipity Point Films	Chris Weitz
170		Zachary Levi	iDeal Partners Film Fund	Ken Kwapis
171		Gabourey Sidibe	Hit the Ground Running Films	Stephen Sommers
172		RenÃ©e Zellweger	An Olive Branch Productions	Anne Fletcher
173		Tom Sturridge	Hyde Park Films	Robert Luketic
174		Ethan Hawke	RabbitBandini Productions	Wes Anderson
175		Tobin Bell	Chad Troutwine Films	Henry Selick
176		Emily Blunt	HighRoad Entertainment	Shawn Levy
177		Kate Beckinsale	Glass Eye Pix	Burr Steers
178		John Cena	Boram Entertainment Inc.	Steve Carr
179		Aamir Khan	26 Films	Richard Kelly
180		Shia LaBeouf	Film4	Mark Neveldine
181		Rain	O2 Filmes	Christian Alvart
182		Vanessa Hudgens	Wild Bunch	Sam Raimi
183		Max Records	New Zealand Film Commission	Alex Proyas
184		Ally Sheedy	Fountainbridge Films	Duncan Jones
185		Jensen Ackles	Bookshop Productions	Hoyt Yeatman
186		Robin Williams	Hobby Films	Tony Scott
187		Sean Patrick Flanery	Beacon Communications	P.J. Hogan
188		Susan Sarandon	Nina Saxon Film Design	Peter Billingsley
189		Carey Mulligan	Franchise Pictures	John Hillcoat
190		Clive Owen	Artisan Entertainment	Nora Ephron

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Master Index	Genre	Actor	Production Company	Director
191		Michael Moore	Lions Gate	Phil Traill
192		Scout Taylor-Compton	Mutual Film Company	David R. Ellis
193		Miley Cyrus	1492 Pictures	Betty Thomas
194		Carter Jenkins	Industry Entertainment	Agnieszka Wojtowicz-Vosloo
195		Josh Stewart	Intermedia Films	Gary Winick
196		Carla Gugino	Warner Bros. Pictures	Lee Daniels
197		Milla Jovovich	NPV Entertainment	Kevin Macdonald
198		Michael Douglas	Conundrum Entertainment	John Hamburg
199		Gemma Arterton	Fine Line Features	Steven Soderbergh
200		Abbie Cornish	Lawrence Bender Productions	Richard Curtis
201		Nia Vardalos	Eureka Pictures	Michael Spierig
202		Derek Mears	Summit Entertainment, LLC	Brandon Camp
203		Bruce Willis	Celador Films	Kevin Greutert
204		BeyoncÃ© Knowles	Rogue Marble	Jean-Marc VallÃ©e
205		Vincent Gallo	Sunswept Entertainment	Dominic Sena
206		James Purefoy	Saturn Films	Renny Harlin
207		Alan van Sprang	Blue Askew	Rajkumar Hirani
208		Michael Jai White	Blueprint Pictures	Grant Heslov
209		David Duchovny	Casey Silver Productions	James McTeigue
210		Nicholas D'Agosto	Playtone	Todd Graff
211		Salman Khan	Virtual Studios	Spike Jonze
212		Alexis Bledel	Senator Entertainment Co	Jorge Blanco
213		Johanna Braddy	Fox Atomic	Todd Solondz
214		Chi Cao	Universal Pictures Corporation	Neil Jordan
215		Natalie Portman	Iyara Films	Patrick Lussier
216		Garrett Hedlund	Warner Bros. Entertainment	Walt Becker
217		Emma Stone	Film Direction	Alejandro AmenÃ¡bar
218		Michael Cera	Starz Media	Troy Duffy
219		Mike Myers	Grosvenor Park Media Ltd.	Tim Blake Nelson
220		Danny Trejo	Lions Gate Films	Lone Scherfig
221		Noah Ringer	Ixtlan Productions	Judd Apatow
222		Patrick Wilson	Carmel Entertainment	Jason Reitman
223		Rick Malambri	Max TV	David Bowers
224		Colin Firth	Zininsa Film Production	Tom Tykwer
225		Jude Law	Likely Story	Justin Lin
226		Keira Knightley	Premise Media Corporation	Michael Moore
227		Julia Roberts	CBS Films	Rob Zombie
228		Ryan Gosling	Appian Way	Peter Chelsom
229		Amy Adams	Fox Searchlight	John Schultz

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230		Jessica Alba	France 3 CinÃ©ma	Marcus Dunstan
231		Katie Holmes	Film 4	Sebastian Gutierrez
232		Jaden Smith	Solaris Film	Olatunde Osunsanmi
233		Zach Galifianakis	Virgin Produced	Brian Koppelman
234		Bill Nighy	TF1 Films Production	J Blakeson
235		John Travolta	Mpower Pictures	Tony Gilroy
236		Keir Gilchrist	Atlas Entertainment	Jane Campion
237		Jackie Earle Haley	Original Film	Greg Mottola
238		Jeffrey Dean Morgan	Dark Horse Entertainment	Mike Judge
239		Josh Brolin	Walt Disney Animation Studios	Oren Moverman
240		Chris Riggi	UK Film Council	Donald Petrie
241		Paul Bettany	Fox 2000 Pictures	Nick Cassavetes
242		Emma Thompson	Battleplan Productions	Joe Wright
243		Kodi Smit-McPhee	Rising Star	Marcus Nispel
244		Madeline Carroll	Benaroya Pictures	Jonathan Mostow
245		Emily Barclay	Starz Animation	Steve Shill
246		Chris Brown	Toledo Productions	Francis Ford Coppola
247		Cher	IM Global	Michael J. Bassett
248		Drew Barrymore	Recorded Pictures Company	George A. Romero
249		Katie Featherston	Mandalay Pictures	Gavin Hood
250		Whitney Able	HW Two	Scott Sanders
251		Michael Fassbender	Cliffjack Motion Pictures	Jonas Elmer
252		Ewan McGregor	120dB Films	Derrick Borte
253		Frances Conroy	Lightstream Pictures	Will Gluck
254		Elisabeth Shue	Hydraulx	Prabhu Deva
255		Rachel McAdams	Neue Road Movies	Bobcat Goldthwait
256		Hilary Swank	Voltage pictures	Vicky Jenson
257		Ashley Bell	Lewis Gilbert Productions	Anders Anderson
258		Naomi Watts	Emperor Motion Pictures	Toby Wilkins
259		Lyndsy Fonseca	FilmDistrict	Bruce Beresford
260		Eric Balfour	Ecosse Films	Tim Burton
261		James Marsden	Ada Films	Pierre Coffin
262		Will Forte	Stardust Pictures	Sylvester Stallone
263		Reese Witherspoon	Mr. Tamborine Man	Martin Scorsese
264		Thomas F. Duffy	Incentive Filmed Entertainment	Joseph Kosinski
265		Alice Eve	Rhombus Media	Nathan Greno
266		Rebecca Hall	Atlas Productions	Lee Unkrich
267		Hayden Panettiere	Chrysalis Films	Florian Henckel von Donnersmarck
268		Dan Aykroyd	EM Media	Mike Newell

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269		Rainn Wilson	Insurge Pictures	Ethan Coen
270		Gwyneth Paltrow	Coatwolf Productions	Louis Leterrier
271		Caitlin Stasey	Bac Films	Jon Turteltaub
272		Chris Messina	Columbia Pictures Corporation	Edgar Wright
273		Andy McAvin	Pandora Pictures	Paul Weitz
274		Paul Giamatti	David Foster Productions	Mike Mitchell
275		Juno Temple	Alliance Atlantis Communications	Joe Carnahan
276		Megan Faccio	Konrad Pictures	Robert Rodriguez
277		Brendan Fraser	Channel Four Films	Dennis Dugan
278		Kevin Spacey	Brownstone Productions	Ben Affleck
279		Brandon Routh	Amblin Entertainment	David Slade
280		James Franco	Anarchos Productions	Tom McGrath
281		Zoe Sloane	Nick Wechsler Productions	Tom Hooper
282		Clifton Collins, Jr.	Scott Rudin Productions	Edward Zwick
283		Danielle Harris	Irish Film Board	Peter Weir
284		Kate Bosworth	Blind Spot Pictures Oy	Miguel Sapochnik
285		Rosanna Arquette	Apatow Productions	Massy Tadjedin
286		Riz Ahmed	Ixtlan	James Mangold
287		Vik Muniz	Plan B Entertainment	Ryan Murphy
288		Kevin Kline	Contrafilm	Derek Cianfrance
289		Hayden Christensen	Wessler Entertainment	Andy Tennant
290		Diane Lane	Dos Corazones	Anand Tucker
291		James Rolleston	Pariah Entertainment Group	Paul Greengrass
292		Edward Norton	Gary Sanchez Productions	Alan Poul
293		Sean Connery	Serenade Films	Josh Gordon
294		Tobey Maguire	Room 101	Garry Marshall
295		Willem Dafoe	Primate Pictures	Troy Nixey
296		James Woods	Journeyman Pictures	Harald Zwart
297		Heather Donahue	Samuel Goldwyn Films	Jonathan Lynn
298		Jim Carrey	France 2 CinÃ©ma	Pierre Morel
299		Elijah Wood	Threshold Entertainment	Anna Boden
300		John Goodman	Newmarket Films	Paul Haggis
301		Michael J. Fox	City Light Films	Samuel Bayer
302		Jeff Goldblum	Boll Kino Beteiligungs GmbH & Co. KG	Sylvain White
303		Ryan Phillippe	Recorded Picture Company (RPC)	David O. Russell
304		Steven Seagal	Harbour Pictures	Nicholas Stoller
305		Method Man	icon	Jimmy Hayward
306		Natasha Henstridge	Epsilon Motion Pictures	Jason Friedberg
307		Chris Rock	Annapurna Pictures	Charles Ferguson

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308		Patricia Arquette	One Race Productions	Scott Stewart
309		Barry Pepper	Constantin Film Produktion	Susanna White
310		Janeane Garofalo	Sidney Kimmel Entertainment	Steve Pink
311		Ben Burtt	21 Laps Entertainment	Matt Reeves
312		Josh Zuckerman	Rainmaker Entertainment	Rob Reiner
313		Clint Eastwood	Di Bonaventura Pictures	Zack Snyder
314		Dev Patel	Paco Cinematografica	John Luessenhop
315		Jeremy Renner	Pathe	Steve Antin
316		Vera Farmiga	3311 Productions	Nanette Burstein
317		Lizzy Caplan	3 Arts Entertainment	Tod Williams
318		Jason Segel	Triggerfish Animation	Mark Romanek
319		Jim Sturgess	Filmworks/FX	Kevin Smith
320		Sean Faris	Patalex IV Productions Limited	Greg Berlanti
321		Ray Stevenson	Tim Burton Animation Company	Neil Marshall
322		John Cho	Aardman Animations	Mike Mills
323		Anna Faris	International Production Company	BjÃ¶rn Stein
324		Emile Hirsch	Warner Independent Pictures (WIP)	John Curran
325		Jackie Chan	Canal Plus	Alexandre Aja
326		Don Cheadle	Revolution Studios	Roger Michell
327		Liv Tyler	Boll KG	Tony Goldwyn
328		Sean Penn	Vulcan Productions Inc.	Daniel Stamm
329		Mark Strong	Bergman Lustig productions	John Carpenter
330		Sean Maguire	Big Beach Productions	Rodrigo CortÃ©s
331		Gabriel Macht	IFC Films	Jay Roach
332		Billy Bob Thornton	Class 5 Films	Lisa Cholodenko
333		Sarah Jessica Parker	Movieworld Productions	Colin Strause
334		Matthew Broderick	Silent Hill DCP Inc.	John Madden
335		Tony Jaa	Thousand Words	Debra Granik
336		Matthew Goode	Sony Pictures Animation	Jorma Taccone
337		Eddie Murphy	Paramount Classics	James L. Brooks
338		David Boreanaz	Cipher Films	Quentin Dupieux
339		Konstantin Khabenskiy	BIM Distribuzione	Jim Field Smith
340		Andy Samberg	Red Hour Films	William Monahan
341		Matt Lanter	Davis Entertainment	Robert Redford
342		Kirk Cameron	Yash Raj Films	Spencer Susser
343		Martijn Lakemeier	Tessalit Productions	Oliver Stone
344		Alan Rickman	Blackfriars Bridge Films	Dan Rush
345		Leonor Varela	Gaylord Films	Anthony Bell
346		Ben Stein	Laura Ziskin Productions	Eric Brevig

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347		Andy Serkis	New Amsterdam Entertainment	Shana Feste
348		Justin Timberlake	Thinkfilm	Shinji Aramaki
349		Larry the Cable Guy	Avenue Pictures Productions	Richard J. Lewis
350		Jamie Bell	Splendid Pictures	Abe Sylvia
351		Kristen Wiig	Whitewater Films	Henry Joost
352		Joel Courtney	AVCO Embassy Pictures	Roger Kumble
353		Emily Browning	Cool Music	George Hickenlooper
354		Amanda Seyfried	Darkwoods Productions	Kevin Munroe
355		Alex Pettyfer	Walden Media	Rob Epstein
356		Jean Dujardin	Pacific Data Images (PDI)	Alex Gibney
357		Tom Hiddleston	Gracie Films	William Dear
358		Asa Butterfield	Entertainment Films	Jim Mickle
359		Anthony Hopkins	Halcyon Pictures	Sngmoo Lee
360		Jim Cummings	MeniThings LLC	Chris Morris
361		Taylor Lautner	Opera Film Produzione	Lucy Walker
362		Anton Yelchin	Greenestreet Films	Shari Springer Berman
363		Zoe Saldana	BBC	Paul W.S. Anderson
364		Emma Bell	Celluloid Dreams	Brad Anderson
365		Michelle Williams	Jigsaw Productions	Taika Waititi
366		Minka Kelly	Twentieth Century-Fox Productions	Lilly Wachowski
367		Kate Hudson	Zoetrope Studios	Frank Darabont
368		Jason Momoa	New Breed Productions	Michael Apted
369		Rowan Atkinson	Orion Pictures	Jon Amiel
370		Neil Patrick Harris	PSO International	John Lasseter
371		Mary Elizabeth Winstead	Blue Parrot Productions	Kimberly Peirce
372		Danny McBride	Lightstorm Entertainment	Stanley Kubrick
373		Ryan Robbins	Walt Disney Productions	Peter Hyams
374		David Arquette	Metro-Goldwyn-Mayer (MGM)	Lasse HallstrÃ¶m
375		Seth Green	PolyGram Filmed Entertainment	Sofia Coppola
376		Nick Frost	Tim Burton Productions	Daniel Myrick
377		Zachary Gordon	Bazmark Films	Anthony Minghella
378		Thomas Horn	Warner Bros. Family Entertainment	MiloÅ¡ Forman
379		Jodie Foster	Kopelson Entertainment	Wayne Wang
380		Patrick Dempsey	A Band Apart	Andrew Adamson
381		Russell Brand	Hollywood Pictures	Gary Trousdale
382		Brendan Gleeson	Jersey Films	Lawrence Guterman
383		Kenny Wormald	Interscope Communications	Peter Howitt
384		Topher Grace	Bel Air Entertainment	Iain Softley
385		Channing Tatum	Shaw Brothers	Luis Mandoki

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386		Sara Paxton	Punch Productions	Andrzej Bartkowiak
387		Ioan Gruffudd	Truth and Soul Pictures Inc	Jesse Dylan
388		AnnaSophia Robb	Handmade Films Ltd.	Bobby Farrelly
389		Robert Pattinson	Great American Films Limited Partnership	Michel Gondry
390		Mandy Moore	Brooksfilms Ltd.	David Wain
391		Woody Harrelson	Python (Monty) Pictures Limited	Andrew Stanton
392		Michael Shannon	Proscenium Films	Steven Spielberg
393		Aimee Teegarden	The Mirisch Corporation	Catherine Hardwicke
394		Regina Advento	Silver Pictures	Marc Forster
395		Alex Kendrick	Propaganda Films	Eric Darnell
396		Portia Doubleday	Compass International Pictures	Timur Bekmambetov
397		Camilla Belle	Kennedy Miller Productions	Danny Boyle
398		Andy Lau	Mandeville Films	Peter Berg
399		Zana MarjanoviÄ‡	Truenorth Productions	Kathryn Bigelow
400		Paula Patton	Skydance Productions	Ben Stiller
401		Kaya Scodelario	Illumination Entertainment	Mark Herman
402		Mark Duplass	Pixar Animation Studios	Gabriele Muccino
403		Krysten Ritter	Danjaq	Chris Williams
404		J.K. Simmons	Universal Pictures International (UPI)	David Frankel
405		Rutger Hauer	Happy Madison Productions	Jeff Wadlow
406		Taylor Schilling	Prime Focus	David Gordon Green
407		Glenn Close	M6 Films	Lexi Alexander
408		Tom Cullen		John Moore
409		Justin Bieber	Babieka	Pete Travis
410		Evan Glodell	Trigger Street Productions	Peter Segal
411		Cindy Baer	Marv Films	Mathieu Kassovitz
412		Ray Romano	No Trace Camping	Martin McDonagh
413		Tommy Lee Jones	Atmosphere Entertainment MM	Jon Hurwitz
414		Adrien Brody	Versus Pictures	Fred Wolf
415		Wesley Snipes	Eurimages	Marco Schnabel
416		Eminem	PeaPie Films	Rob Minkoff
417		Parminder Nagra	Ram Bergman Productions	Jeffrey Nachmanoff
418		Hugh Grant	Mad Chance	Gregory Hoblit
419		Guy Pearce	Forward Pass	Bryan Bertino
420		Jonathan Taylor Thomas	RT Features	Gus Van Sant
421		Brian O'Halloran	Chernin Entertainment	Gil Kenan
422		Melanie Lynskey	RatPac-Dune Entertainment	David Moreau
423		Marcello Mastroianni	Fox International Productions	Oxide Pang Chun
424		Jamie Foxx	Cross Creek Pictures	Gregor Jordan

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425		Ian McKellen	Twentieth Century Fox Animation	Michael Patrick King
426		Noomi Rapace	Point Grey Pictures	Peter Cattaneo
427		Kelly Macdonald	Dragonfly Entertainment	Robert Stevenhagen
428		Suraj Sharma	Groundswell Productions	Gina Prince-Bythewood
429		John Leguizamo	Sony Pictures Entertainment	Tony Jaa
430		John C. Reilly	Senator Film	Gavin O'Connor
431		Anna Kendrick	Grindstone Entertainment Group	Michael McCullers
432		Logan Lerman	Blue-Tongue Films	Julian Jarrold
433		Jason Biggs	Orange Studios	Rian Johnson
434		Kristen Connolly	Film Art Media	Brian Robbins
435		Chris Pine	Nuyorican Productions	Dave Bullock
436		Karl Urban	Oops Doughnuts Productions	Andrey Kravchuk
437		Taylor Kitsch	Blinding Edge Pictures	Kirk De Micco
438		Jessica Chastain	Living Films	Howard Deutch
439		Gina Carano	Revolt Films	RyÃhei Kitamura
440		Roselyn SÃ¡nchez	Origin Pictures	Martin Koolhoven
441		Sacha Baron Cohen	MTV Films	Randall Miller
442		Julia Dietze	Gran Via Productions	Alex Rivera
443		Charlie Tahan	Nickelodeon Movies	Nathan Frankowski
444		Danielle Panabaker	TriStar Productions	Joe Johnston
445		Sean Hayes	QED International	Kenneth Branagh
446		Eva Longoria	Haven Entertainment	Andrew Niccol
447		Bryan Cranston	Affirm Films	Simon West
448		Sheri Moon Zombie	Mel's Cite du Cinema	Jennifer Yuh
449		Kirsten Dunst	Moving Picture Company (MPC)	Seth Gordon
450		Fernanda Andrade	Color Force	Jake Kasdan
451		Eric Bana	Bristol Automotive	Martin Campbell
452		QuvenzhanÃ© Wallis	Red Wagon Entertainment	Ivan Reitman
453		Ashley Greene	Donners' Company	Bill Condon
454		Frank Langella	Sierra / Affinity	Terrence Malick
455		Charlie Sheen	Marvel Enterprises	D.J. Caruso
456		Katy Perry	Malpaso Productions	Michel Hazanavicius
457		Uma Thurman	Roth Films	Frank Coraci
458		Albert Brooks	Mad Hatter Entertainment	Gore Verbinski
459		Luke Wilson	Marc Platt Productions	Brad Furman
460		Halle Berry	Hollywood Gang Productions	Mikael HÃ¥fstrÃ¶m
461		Jonathan Cherry	We're Not Brothers Productions	Stephen Anderson
462		Sid Haig	Aldamisa Entertainment	John Singleton
463		Charlotte Rampling	Imagenation Abu Dhabi FZ	Jaume Collet-Serra

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464		Luke Kirby	Offspring Entertainment	George Nolfi
465		Bill Paxton	Troublemaker Studios	Drake Doremus
466		Alex D. Linz	Crescendo Productions	Tom Hanks
467		Henry Cavill	Low Spark Films	Steven Quale
468		Lily Collins	DisneyToon Studios	Mark Mylod
469		Jane Levy	The Safran Company	Cameron Crowe
470		ChloÃ Grace Moretz	Screen Australia	Chris Gorak
471		Nicholas Hoult	Weed Road Pictures	Tate Taylor
472		Marlon Wayans	RatPac Entertainment	Bennett Miller
473		Rooney Mara	Altitude Film Entertainment	Christian E. Christiansen
474		Geoffrey Rush	Warner Bros. Interactive Entertainment	Luke Greenfield
475		Charlie Rowe	Warner Bros. Animation	Oliver Parker
476		Luke Evans	Zanuck Independent	Todd Strauss-Schulson
477		Ashton Kutcher	Unified Pictures	Matthijs van Heijningen Jr.
478		Lake Bell	Winkler Films	Brett Ratner
479		Aubrey Plaza	Heyday Films	Gonzalo LÃ³pez-Gallego
480		Paul Walker	The Halcyon Company	Wes Craven
481		Jake T. Austin	Watermark	Simon Wells
482		Forest Whitaker	Green Hat Films	Joe Cornish
483		John Jarratt	Left Tackle Pictures	Stephen Daldry
484		Brit Marling	Maverick Films	Jodie Foster
485		Lindsay Lohan	Flower Films (II)	Rod Lurie
486		Zhang Ziyi	Laika Entertainment	Raja Gosnell
487		Zach Braff	ImageMovers	Kelly Asbury
488		Orlando Bloom	Lin Pictures	Elliott Lester
489		Heath Ledger	Mystery Clock Cinema	Alexander Payne
490		Antonio Banderas	Lunar Industries	Tim Hill
491		Jennifer Garner	Wild West Picture Show Productions	John Michael McDonagh
492		Peter Sallis	Practical Pictures	Jesse Peretz
493		Jay Hernandez	Harbor Light Entertainment	Matthew Chapman
494		Robert Redford	The Montecito Picture Company	Craig Brewer
495		Johnny Knoxville	Studio Canal	Roman Polanski
496		David Strathairn	Furst Films	Michael Dowse
497		Jet Li	Stuber Productions	James Bobin
498		Al Pacino	Infinity Features Entertainment	David Schwimmer
499		Heike Makatsch	Twisted Pictures	Jason Winer
500		Tom Welling	Dark Castle Entertainment	Madonna
501		Kristanna Loken	Smoke House	Ti West
502		Philip Seymour Hoffman	Goldsmith-Thomas Productions	Alister Grierson

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503		Michael Angarano	Ilion Animation Studios	Douglas McGrath
504		Felicity Huffman	Tapestry Films	David Cronenberg
505		Emily Deschanel	Telecinco Cinema	Sean McNamara
506		Anthony Rapp	Stage 6 Films	Mike Disa
507		Matthew Le Nevez	Grand Army Entertainment	Jim Sheridan
508		Jenna Dewan	Spectacle Entertainment Group	Dermot Mulroney
509		Presley Chweneyagae	It's a Laugh Productions	Jeff Nichols
510		Ben Whishaw	LD Entertainment	Joe Nussbaum
511		Radha Mitchell	Saga Film	Wim Wenders
512		Kiefer Sutherland	CinemaNX	William Friedkin
513		Ed Speleers	Laura Bickford Productions	John Whitesell
514		Aaron Stanford	PathÃ© Renn Productions	Angel Gracia
515		Kurt Russell	Czech Anglo Productions	Benny Chan
516		Liev Schreiber	Artfire Films	Salim Akil
517		Ken Watanabe	Six Point Harness	Andrea Arnold
518		Martin Lawrence	Edmonds Entertainment Group (EEG)	Lynn Shelton
519		Lucas Black	Premiere Picture	Kat Coiro
520		Tim Allen	Moving Pictures, DPI	Phyllida Lloyd
521		Emma Roberts	Jerkschool Productions	Jim Kohlberg
522		Judi Dench	Cold Spring Pictures	Jason Eisener
523		Aml Ameen	Boy in the Box	Paul Johansson
524		Sarah Michelle Gellar	Ozla Pictures	Rodrigo GarcÃa
525		Hrithik Roshan	Team Todd	Andrew Haigh
526		Billy Connolly	LivePlanet	Evan Glodell
527		James Blackwell	Olive Bridge Entertainment	Chris Wedge
528		Samuel Anderson	Peninsula Films	Barry Sonnenfeld
529		Al Gore	Dune Entertainment III	Curtis Hanson
530		Thomas Turgoose	Red Om Films	Gurinder Chadha
531		Jamel Debbouze	Silverwood Films	Adam Shankman
532		Keke Palmer	Bona Fide Productions	Andrew Davis
533		Craig T. Nelson	Karz Entertainment	Chuck Russell
534		Ron Perlman	Jerry Weintraub Productions	Phil Alden Robinson
535		Hilary Duff	FidÃ©litÃ© Films	Tom Dey
536		Sanaa Lathan	Platinum Dunes	Roger Allers
537		David Belle	Sony Pictures Home Entertainment	Robert Altman
538		Thomas Jane	Three Strange Angels	MÃ¥ns MÃ¥rlind
539		Sarah Polley	EFTI	Mark Andrews
540		Richard Gere	Battle Mountain Films	Rupert Sanders
541		Shane Carruth	Gold Circle Films	Ang Lee

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542		Randy Quaid	Mimran Schur Pictures	Seth MacFarlane
543		Trey Parker	Goldcrest Pictures	Juan Antonio Bayona
544		Jim Caviezel	Pantheon Entertainment Corporation	Rich Moore
545		LL Cool J	Versus Entertainment	Gary Ross
546		Frankie Muniz	10th Hole Productions	Len Wiseman
547		Tracey Ullman	Winter's Bone Productions	Stephen Chbosky
548		Rory Culkin	Backup Films	Drew Goddard
549		Marlon Brando	Mosaic	Peter Ramsey
550		Adrienne Barbeau	Henceforth	Boaz Yakin
551		Patton Oswalt	American Film Company	Mike McCoy
552		Claire Danes	The Last Picture Company	Genndy Tartakovsky
553		Ellen Page	Ambush Entertainment	Kirk Jones
554		Josh Hutcherson	Ambience Entertainment	Larry Charles
555		Jason Lee	Robert Simonds Productions	Timo Vuorensola
556		Jerry Seinfeld	Rollercoaster Entertainment	Peter Lord
557		Dan Castellaneta	platinum studios	Stephen St. Leger
558		Steven Pasquale	Green Film Company	James Watkins
559		Josh Hartnett	Prelude Pictures	Daniel Espinosa
560		Glen Hansard	Belladonna Productions	Asger Leth
561		Freddie Highmore	Drafthouse Films	Henry Alex Rubin
562		Ray Winstone	Almega Projects	John Gulager
563		Catherine Zeta-Jones	Herrick Entertainment	Andrew Dominik
564		Timothy Olyphant	Fast Track Productions	Dan Bradley
565		Simon Pegg	Unison Films	Julie Anne Robinson
566		Jack Nicholson	Groucho II Film Partnership	Sam Fell
567		Malcolm McDowell	Jinks/Cohen Company	Dean Wright
568		Cuba Gooding Jr.	Alphaville Films	Anthony Hemingway
569		Matthew Macfadyen	Notting Hill Pictures	Leslye Headland
570		Omar Metwally	The Independent Film Channel Productions	Brian Klugman
571		Meg Ryan	Pole Star	Walter Salles
572		Asia Argento	FilmColony	William Brent Bell
573		Khalid Abdalla	Brood Syndicate	David Palmer
574		Casey Affleck	Haxan Films	Stefan Ruzowitzky
575		Javier Bardem	View Askew Productions	Benh Zeitlin
576		Patrick Stewart	Walt Disney Feature Animation	Todd Lincoln
577		Ben Kingsley	Lawrence Gordon Productions	Jake Schreier
578		Ryan Kwanten	Storm King Productions	Lawrence Kasanoff
579		Neil Dudgeon	Senator Film Produktion	Dan Cutforth
580		Gaspard Ulliel	Alloy Entertainment	Tom Shadyac

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581		Brian Keith Allen	Hypnotic	Mark Steven Johnson
582		Bruce Campbell	Kingsgate Films	Terry Zwigoff
583		Sigourney Weaver	Cruise/Wagner Productions	Richard Linklater
584		Martin Sheen	Internationale Filmproduktion Blackbird Erste	Clark Johnson
585		Angus Scrimm	MHF Zweite Academy Film	Lawrence Kasdan
586		Matt Dillon	Firm Films	Patty Jenkins
587		Stephen Baldwin	Scion Films	Martin Brest
588		Jason David Frank	China Film Co-Production Corporation	Uwe Boll
589		Ice Cube	Cohen/Pearl Productions	David Mackenzie
590		Ralph Fiennes	VN Productions	FranÃ§ois Ozon
591		Scott Weinger	Initial Entertainment Group	Nigel Cole
592		Stephen Rea	New Regency Pictures	Ã‰mile Gaudreault
593		Chevy Chase	Darren Star Productions	Roger Spottiswoode
594		Mickey Rooney	Michaels-Goldwyn	Chris Buck
595		Jamie Lee Curtis	Linson Entertainment	Alfonso CuarÃ³n
596		Laurence Olivier	LionsGate	Shane Black
597		William H. Macy	New Wave Entertainment	David Twohy
598		Michael Jordan	Sherazade Film Development	Thor Freudenthal
599		James Cromwell	Prime Time	Denis Villeneuve
600		Jada Pinkett Smith	Intellectual Properties Worldwide	Tommy Wirkola
601		Madonna	Marvel Entertainment	Harmony Korine
602		Miriam Margolyes	Vendome Pictures	Fede Alvarez
603		Mark Addy	Rat Entertainment	Seth Rogen
604		Chris O'Donnell	Mosaic Media Group	Chan-wook Park
605		Susan Ward	Reliance BIG Entertainment	James Ponsoldt
606		Henry Thomas	La Petite Reine	David Soren
607		Dustin Hoffman	Infinitum Nihil	Steven Knight
608		Sean Gullette	Stars Road Entertainment	Andy Muschietti
609		Jason Flemyng	Mark Canton Productions	Callan Brunker
610		Timothy Dalton	Andrea Sperling Productions	Michael Tiddes
611		Paige O'Hara	Playtone Productions	Giuseppe Tornatore
612		Roddy Piper	Big Talk Productions	Allen Hughes
613		James Le Gros	1984 Private Defense Contractors	Neil Nightingale
614		Jennifer Grey	Wild Ocean Films	Joshua Michael Stern
615		Mel Brooks	Image Entertainment	Lake Bell
616		Donald Pleasence	Neunte Babelsberg Film	Maggie Carey
617		Roy Scheider	Before the Door Pictures	Anthony Silverston
618		Graham Chapman	Current Entertainment	Kasi Lemmons
619		Tim Curry	Foresight Unlimited	Greg McLean

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620		John Lennon	Zanagar Films	Zal Batmanglij
621		Robert Duvall	Crystal City Entertainment	Paul Schrader
622		Timothy Bottoms	Semtex Films	John Glen
623		Alec Baldwin	Dark Sky Films	Mark Dindal
624		Macaulay Culkin	Wayfare Entertainment	Rob Bowman
625		James Cagney	The Movie Network	Steve Box
626		Sam Neill	Morgan Creek Productions	Cory Edwards
627		Janet Jackson	Waypoint Entertainment	Sydney Pollack
628		Linda Blair	REI Capital	Jay Chandrasekhar
629		George Lazenby	Rickshaw Productions	Stephen Gaghan
630		John Belushi	Zweites Deutsches Fernsehen (ZDF)	Dean Parisot
631		Bryce Dallas Howard	Arc Productions	David Ayer
632		Theo James	China Film Group Corporation (CFGC)	Jim Jarmusch
633		Phyllis Smith	Film4 Productions	Rupert Wainwright
634		Alicia Vikander	Next Wednesday Productions	Liev Schreiber
635		Alexandra Daddario	Dot Dot Dot Productions	Duncan Tucker
636		Jamie Dornan	Essential Pictures	Stephen Kay
637		Michael Keaton	Yer Dead Productions	Brett Leonard
638		Dylan Minnette	The Bureau	Jeremy Haft
639		Jacob Tremblay	Magical Elves Productions	Niki Caro
640		Jack Bright	YouTube	David Jacobson
641		Corey Hawkins	Kennedy/Marshall Company, The	Spike Lee
642		Rose Leslie	Canal+Polska	Christophe Gans
643		Michiel Huisman	Marvel Entertainment, LLC	Steve 'Spaz' Williams
644		Haley Bennett	Initial Entertainment Group (IEG)	Stefen Fangmeier
645		Rihanna	Hofflund/Polone	Alejandro Agresti
646		Luke Bracey	Mikona Productions GmbH & Co. KG	John A. Davis
647		Sienna Miller	Kintop Pictures	Robert De Niro
648		Leehom Wang	Eric's Boy	Steven Zaillian
649		Kate Mara	Kalima Productions GmbH & Co. KG	Joe Roth
650		Ralph Ineson	TiMe Film- und TV-Produktions GmbH	Jared Hess
651		Margot Robbie	Mace Neufeld Productions	Wolfgang Petersen
652		Zachary Quinto	Beverly Detroit	Steven Shainberg
653		Benjamin Walker	Dentsu	Anthony Russo
654		Bill Melendez	Media Magik Entertainment	Jonathan Dayton
655		Bitsie Tulloch	X-Filme Creative Pool	Keenen Ivory Wayans
656		Lorenza Izzo	Mr. Mudd Production	Jim Sonzero
657		Diane Keaton	Extension 765	Elizabeth Allen Rosenbaum
658		Abigail Breslin	Mutant Enemy Productions	Woody Allen

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659		Liam Hemsworth	Indian Paintbrush	Richard Eyre
660		Robbie Amell	Bandito Brothers	Menhaj Huda
661		James Ransone	27 Films Production	Stephen Frears
662		Logan Miller	Corner Store Entertainment	Wayne Kramer
663		Ed Oxenbould	Di Bonaventura Picture	Susannah Grant
664		Toni Collette	Liddell Entertainment	Takashi Shimizu
665		Kiersey Clemons	Lucasfilm Ltd.	Liam Lynch
666		Virginia Gardner	BCDF Pictures	Sanjay Gadhvi
667		Bill Fagerbakke	Prototype	Andrew Currie
668		Stefanie Scott	Kim and Jim Productions	Davis Guggenheim
669		Alex Jennings	Court 13 Pictures	Shane Meadows
670		Zooey Deschanel	TÃ©lÃ©film Canada	Rachid Bouchareb
671		Max Greenfield	Laurinfilm	Doug Atchison
672		Ryan Shoos	Dimension Films (US version)	Mark Waters
673		T.C. Stallings	Newman/Tooley Films	David S. Goyer
674		Jonny Beauchamp	Y Productions	Kerry Conran
675		Michael Nyqvist	Brightlight Pictures	Mark Rosman
676		Ryan Potter	Hughes Entertainment	Jonathan Hensleigh
677		Ki Hong Lee	South Side Amusement Company	Shane Carruth
678		Rosamund Pike	Across the River Productions	Will Finn
679		Rene Russo	Radar Pictures	Terry George
680		Stanley Tucci	Chartoff Productions	Trey Parker
681		Joel Edgerton	Madhouse Entertainment	Mel Gibson
682		Elle Fanning	Don Carmody Productions	Taylor Hackford
683		Elizabeth Banks	Division Films	Kevin Allen
684		Eva Green	WWE Studios	John Waters
685		Peter Capaldi	Electric City Entertainment	Jacob Aaron Estes
686		Chad Michael Murray	Andrew Lauren Productions (ALP)	Irvin Kershner
687		Jennifer Connelly	Toma 78	Sean Penn
688		Briana Evigan	Mainframe Entertainment	GÃ¡bor CsupÃ³
689		Laurence Fishburne	Wayans Bros. Entertainment	Antoine Fuqua
690		Ed Harris	Inferno International	Chris Miller
691		Taraji P. Henson	Dillywood	Steve Hickner
692		Ben Feldman	In A World	Joel Schumacher
693		Alfre Woodard	Prettybird	Kevin Lima
694		Yvonne Strahovski	Alberta Film Entertainment	David Silverman
695		Noah Wiseman	Will Vinton Studios	Mike Nichols
696		Jessica Brown Findlay	Raw Nerve	John Carney
697		Olga Kurylenko	Kanbar Entertainment	Kirsten Sheridan

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698		Onni Tommila	Gerber Pictures	Oren Peli
699		Jason Dohring	Section Eight Productions	Darren Lynn Bousman
700		Scott Speedman	Debra Hill Productions	Michael Davis
701		Alan Tudyk	Cooper's Town Productions	Scott Hicks
702		David Thewlis	Gunn Films	Ash Brannon
703		Helen Hunt	Stillking Films	Xavier Gens
704		Ashley Judd	Senator International	Andy Fickman
705		Kelly Reilly	City Lights Pictures	Chris Sivertson
706		Johnny Flynn	Industrial Development Corporation of South Africa	Michael Haneke
707		Sean Astin	Media Asia Films	Fred Savage
708		Shane Harper	Syncopy	Russell Mulcahy
709		Barbara Bach	Four By Two Productions	Michael Winterbottom
710		MÃ©lanie Laurent	Nouvelles Ã‰ditions de Films (NEF)	Frank Oz
711		Rupert Grint	Major Studio Partners	Stephen Hopkins
712		Christopher Plummer	Next Entertainment	Mark Helfrich
713		Jason Cope	Edward R. Pressman Film	Jon Kasdan
714		Ed Helms	Kaplan/Perrone Entertainment	Aristomenis Tsirbas
715		Amanda Peet	11:11 Mediaworks	Richard LaGravenese
716		Bruno Campos	Persistent Entertainment	Dario Argento
717		Quinton Aaron	Storefront Pictures	Peter Hedges
718		Teri Hatcher	Stealth Films Limited	Scott Frank
719		Leslie Mann	Granada Film Productions	Bernie Goldmann
720		Keir O'Donnell	True Grit Productions	Susanne Bier
721		Amy Smart	Astral Media	Mike Binder
722		Justin Long	The Kennedy/Marshall Company	Alastair Fothergill
723		Rose Byrne	Laurence Mark Productions	John Dahl
724		PenÃ©lope Cruz	Golden Mean	Garth Jennings
725		Hugh Dancy	Amen Ra Films	Rod Hardy
726		Michael C. Hall	Natural Nylon Entertainment	Peter Webber
727		Malin Ã…kerman	Yari Film Group	Bruce Campbell
728		Thomas Haden Church	VIP 3 Medienfonds	Don Coscarelli
729		Shantel VanSanten	Detour Film Production	Richard Marquand
730		David Cross	Michael London Productions	John Badham
731		Mo'Nique	FilmEngine	Mike Figgis
732		Rashida Jones	Alfran Productions	Bryan Spicer
733		Scott Bakula	EDI	Paul Verhoeven
734		Jodelle Ferland	Hal Lieberman Company	Ted Berman
735		Costas Mandylor	DreamWorks Pictures	Rick Rosenthal
736		Jim Broadbent	Participant Productions	Desmond Davis

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737		Ashley Scott	Samson Films	Brian De Palma
738		Kareena Kapoor	CJ Entertainment	Joel Coen
739		Naomie Harris	Solana Films	Baz Luhrmann
740		Aly Michalka	Angry Films	Joe Pytka
741		Catherine Keener	Mayhem Pictures	Jonathan Frakes
742		Seann William Scott	Material Entertainment	Alan Parker
743		Paul Reubens	Storyline Entertainment	John Woo
744		Alicja Bachleda-CuruÅ	360 Pictures	Brian Levant
745		Jaime King	2 Entertain	Mary Lambert
746		Max Minghella	Kinematograf Wien	Stephen Norrington
747		Norman Reedus	Revolution Films	Tony Kaye
748		Richard Dreyfuss	Parabolic Pictures	Stephen Herek
749		Olivia Williams	Level 1 Entertainment	Simon Wincer
750		Thora Birch	Land Films Inc.	Emile Ardolino
751		Billy Ray Cyrus	Medusa Film	Mel Brooks
752		Austin Butler	MacDonald/Parkes Productions	Guy Hamilton
753		Michael Reilly Burke	Mirage Enterprises	Jim Sharman
754		Adrianne Palicki	The Ladd Company	Terence Young
755		Will Patton	Eclipse Catering	Richard Lester
756		Martin Compston	Neal Street Productions	Don Siegel
757		Jared Padalecki	Bi-polar Pictures	Peter Bogdanovich
758		Maribel VerdÃº	Hammer & Tongs	John McTiernan
759		Pete Postlethwaite	Australian Film Finance Corporation	Billy Wilder
760		Kenneth Welsh	ETIC Films	Marco Brambilla
761		Arsenio Hall	Brandywine Productions	Uli Edel
762		Harry Connick Jr.	Pacific Western	Peter R. Hunt
763		Demi Moore	Delphi Films	John Landis
764		Eric Christian Olsen	Delphi II Productions	
765		Ayesha Takia	TaliaFilm II Productions	
766		Alexie Gilmore	Sherwood Productions	
767		Zach Gilford	Juno Pix	
768		Josh Lucas	Bad Hat Harry Productions	
769		Gil McKinney	Intial Productions	
770		Chengwu Guo	Alcor Films	
771		Emma Watson	Goldcrest Films International	
772		Hailee Steinfeld	Peerford Ltd.	
773		Michelle Rodriguez	Northern Lights Entertainment	
774		Salma Hayek	Rysher Entertainment	
775		Nicola Peltz	Cinergi Pictures Entertainment	

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776		Adam G. Sevani	Allied Filmmakers	
777		Dejan Angelov	Haft Entertainment	
778		Alex O'Loughlin	Newmarket Capital Group	
779		Jessica Biel	Polygram Filmed Entertainment	
780		Jonathan Rhys Meyers	Global Entertainment Productions GmbH & Company Medien KG	
781		Katie Cassidy	Turman-Foster Company	
782		John Malkovich	The Turman-Morrissey Company	
783		William Ackman	Protozoa Pictures	
784		Callan McAuliffe	Krisjair	
785		Christina Aguilera	Alive Films	
786		David Bierend	Spacegate Productions	
787		Tracy Morgan	Vestron Pictures	
788		Scoot McNairy	Carolco Pictures	
789		Josh Duhamel	Zanuck/Brown Productions	
790		Minnie Driver	Michael White Productions	
791		Patrick Fabian	Walter Shenson Films	
792		Amber Heard	Malpaso Company	
793		JosÃ© Luis GarcÃa PÃ©rez	BBS Productions	
794		Annette Bening	Caravan Pictures	
795		Scottie Thompson	Kouf/Bigelow Productions	
796		John Hawkes	TAFT Entertainment Pictures	
797		Nick Nolte	Hoya Productions	
798		David Bowe	Falcon International Productions	
799		Rachel Hurd-Wood	Oregon Film Factory	
800		Jacob Vargas	CatchPlay	
801			Mid Atlantic Films	
802		Jeremy Dozier	Walt Disney Studios Motion Pictures	
803		Melody C. Roscher	B24	
804		Ken Jeong	Revolution Sun Studios	
805		Kelly Preston	A113	
806		Peter Stormare	Michael De Luca Productions	
807		Jade Viggiano	China Movie Channel	
808		Cheech Marin	TSG Entertainment	
809		Kelly McGillis	Sony Pictures Entertainment (SPE)	
810		Jang Dong-gun	Scholastic Entertainment	
811		Diane Kruger	A24	
812		Nigel Lindsay	Buena Vista	
813		Paul Dano	Item 7	
814		Wentworth Miller	One Race Films	

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815		Thandie Newton	Sierra/Affinity	
816		Craig Hall	Lemming Film	
817		Michael Clarke Duncan	Number 9 Films	
818		Sophie Marceau	DMG Entertainment	
819		ChloÃ Sevigny	OddLot Entertainment	
820		Gabriel Byrne	Rooks Nest Entertainment	
821		Kathleen Turner	Feigco Entertainment	
822		Michael C. Williams	Zaftig Films	
823		Courtney Love	Infinite Frameworks Studios	
824		Billy Crystal	Anton Capital Entertainment (ACE)	
825		Corey Burton	LStar Capital	
826		Elizabeth Perkins	Good Universe	
827		Rachael Leigh Cook	Freerunning	
828		DMX	Silver Reel	
829		Redman	STX Entertainment	
830		Tim Robbins	CityMation	
831		Andy Davoli	Apollo Media	
832		David Hyde Pierce	Vast Entertainment	
833		Michael Caine	Broken Road Productions	
834		Elissa Knight	Exclusive Media Group	
835		Terrence Howard	Brothers Dowdle Productions	
836		Amanda Crew	IamOTHER Entertainment	
837		Christopher Carley	Nickelodeon Animation Studios	
838		Freida Pinto	High Frequency Entertainment	
839		Anthony Mackie	Dune Films	
840		Julie Benz	Chapter One Films	
841		Joan Allen	Management 360	
842		Rosario Dawson	Hasbro Studios	
843		Jessica Lucas	Kendrick Brothers Productions	
844		Sienna Guillory	Bulletproof Cupid	
845		Marisa Tomei	87Eleven	
846		Dominic West	3Foot7	
847		Matthew Fox	Right of Way Films	
848		Michelle Yeoh	Avi Arad Productions	
849		Jeffrey Donovan	Viz Media	
850		Kal Penn	SPD Films	
851		Christina Ricci	Outlaw Sinema	
852		John Turturro	Lucamar Productions	
853		Billy Burke	Tonik Productions	

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854		Harry Treadaway	Cinelou Films	
855		Tom Wilkinson	Kilburn Media	
856		Alessandro Nivola	FilmNation Entertainment	
857		Shahkrit Yamnarm	Big Kid Pictures	
858		Carmen Electra	AR Films	
859		Kim Basinger	Jolie Pas	
860		Kim Cattrall	Prana Studios	
861		Josh Gad	Evergreen Media Group	
862		Queen Latifah	Smoking Gun Productions	
863		Sarunyu Wongkrachang	Seven West Media	
864		Jon Voight	Jim Henson Company, The	
865		Elizaveta Boyarskaya	Film House Germany	
866		Cheryl Hines	Rob Thomas Productions	
867		Vanessa Lachey	DC Entertainment	
868		Vinnie Jones	Voltage Pictures	
869		Erin Bethea	Decoding Annie Parker	
870		Melody Klaver	Boxing Cat Films	
871		Richard Dawkins	Zehnte Babelsberg	
872		Hugo Weaving	Key Creatives	
873		Karin Konoval	Panorama Films	
874		Dakota Goyo	Zucker/Netter Productions	
875		Jason Sudeikis	Imprint Entertainment	
876		BÃ©rÃ©nice Bejo	Internationale Filmproduktion Blackswan	
877		CiarÃ¡n Hinds	Kurtzman/Orci	
878		Colin O'Donoghue	Radical Media	
879		Travis Oates	Xingu Films	
880		Cam Gigandet	Whamaphram Productions	
881		Jennifer Morrison	Easy There Tiger Productions	
882		Kellan Lutz	Fortis Films	
883		Felicity Jones	Parallel Zide	
884		Cliff Curtis	Bagdasarian Productions	
885		RamÃ³n RodrÃguez	Lleju Productions	
886		Eddie Redmayne	Andell Entertainment	
887		Rachael Taylor	Misher Films	
888		Viola Davis	Pictures in Paradise	
889		Leighton Meester	Camp / Thompson Pictures	
890		Jay Chou	20th Century Fox Home Entertainment	
891		Ginnifer Goodwin	Revelations Entertainment	
892		Rose McGowan	Hand Made Films	

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893		Warren Christie	Mod Producciones	
894		Neve Campbell	Tezuka Production Company Ltd.	
895		Joan Cusack	Trancas International Films	
896		Jodie Whittaker	Millar Gough Ink	
897		Devon Bostick	Imaginarium Entertainment Group	
898		Mel Gibson	Focus Films	
899		Jayma Mays	Paul Schiff Productions	
900		Paddy Considine	Curmudgeon Films	
901		Julianne Hough	Wintergreen Productions	
902		Sprague Grayden	Tornasol Films	
903		Andrea Riseborough	Wandering Star Pictures	
904		Pat Healy	New Romero	
905		Henry Hopper	Ars Nova	
906		Armie Hammer	Epidemic Pictures	
907		Richard Roxburgh	2 Bridges Productions	
908		Steve Buscemi	Phoenix Pictures	
909		Thomas McDonell	Mike Zoss Productions	
910		Malou Airaudo	Everyman Pictures	
911		Ken Bevel	Bedlam Productions	
912		Alexa PenaVega	Exclusive Films	
913		Nicholas Tse	Star Entertainment	
914		Goran KostiÄ‡	Rice Films	
915		Julie Bowen	Serendipity Productions	
916		James Northcote	China Film Group	
917		Alex Shaffer	Magic Light Pictures	
918		Anthony Stewart Head	Hwy61	
919		Lou Taylor Pucci	Aversano Films	
920		Chris Carmack	New Crime Productions	
921		Gregory Smith	Animal Logic	
922		Paul Johansson	Village Roadshow Pictures Asia	
923		Chris New	Maraci/Edelstein Films	
924		Jessie Wiseman	Chako Film Company	
925		Moica	Innocence Productions	
926		Franka Potente	Arcade Pictures	
927		Thomas Kretschmann	A Bigger Boat	
928		Kris Kristofferson	Gilbert Films	
929		Daniel Day-Lewis	Pioneer Pictures	
930		Mekhi Phifer	Road Rebel	
931		Shane West	Arte France	

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932		Francesca Neri	Projection Pictures	
933		Kelly Hu	Crest Animation Productions	
934		George H. W. Bush	Crimson Bolt	
935		Samantha Mumba	Night Chronicles	
936		Christina Applegate	Vortex Words Pictures	
937		Rachael Harris	Long Distance Films	
938		Jeff Anderson	Human Worldwide	
939		Andie MacDowell	Lone Runner Entertainment	
940		Sophia Loren	Off Hollywood Pictures	
941		Julie Walters	Warp Films	
942		Irrfan Khan	Tax Credit Finance	
943		Sarah Silverman	Whenua Films	
944		Skylar Astin	Taurus Film	
945		Michelle Pfeiffer	Stanley Kubrick Productions	
946		Alyson Hannigan	Miramax	
947		Chris Sarandon	B.D.S. Productions Inc.	
948		Olivia Thirlby	The Magnolia Project	
949		Lynn Collins	Timnick Films	
950		Emilio Rivera	The Saul Zaentz Company	
951		Sayed Badreya	Native Pictures Productions	
952		Christopher Kirby	Animationwerks	
953		Alison Brie	Baumgarten Management and Productions (BMP)	
954		Maggie Grace	Marvel Knights	
955		Winona Ryder	Collision Entertainment	
956		John Krasinski	Kanzaman	
957		Hope Davis	Internationale Filmproduktion	
958		Matt Bush	Huayi Brothers	
959		Will Sasso	Sony Pictures Releasing	
960		Allison Janney	Toff Guy Films	
961		Tucker Albrizzi	Odd Lot Entertainment	
962		Andy GarcÃa	Home Box Office (HBO)	
963		David Oyelowo	Avery Pix	
964		Bruce Davison	HanWay Films	
965		Simon Quarterman	Guy Walks into a Bar Productions	
966		Dax Shepard	Odyssey Entertainment	
967		Dwight Henry	Art Linson Productions	
968		Sebastian Stan	3 in the Box	
969		Sarah Gadon	GreeneStreet Films	
970		Wayne Brady	Global Entertainment Group	

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971		Lucy Liu	Isabella Films B.V.	
972		Ellen DeGeneres	Chartoff-Winkler Productions	
973		Gregor Fisher	Film Rites	
974		Tony Cox	Reliance Entertainment	
975		Larenz Tate	Lennox House Films	
976		David McCullough	Random Films	
977		Tyron Leitso	Total Entertainment	
978		Bill Moseley	Bay Films	
979		Tilda Swinton	Buckaroo Entertainment	
980		Ludivine Sagnier	Gambit Pictures	
981		Ginette Reno	CinÃ©CinÃ©ma	
982		John Broadwater	Vinyl Films	
983		Jonathan Pryce	2S Films	
984		Olek Krupa	Paradox Entertainment	
985		Joey Lauren Adams	Storefront Films	
986		Harvey Keitel	Bekmambetov Projects Ltd.	
987		Idina Menzel	Dr D Studios	
988		Charlie Hunnam	Kerner Entertainment Company	
989		Jai Courtney	Washington Square Films	
990		Mireille Enos	Ad Hominem Enterprises	
991		Jamie Campbell Bower	VIP Media Group	
992		Selena Gomez	Hasbro	
993		Eleanor Tomlinson	Dark Harbor Stories	
994		Agata Buzek	Sanctum Australia	
995		Nikolaj Coster-Waldau	Millbrook Pictures	
996		Rob Corddry	Brookwell-McNamara Entertainment	
997		Essence Atkins	Voodoo Production Services	
998		Adelaide Clemens	TheThird Mind Pictures	
999		Tao Okamoto	Grove Hill Productions	
999		Tao Okamoto	Grove Hill Productions	
1000		Fred Melamed	ARTE	
1001		Johnny Simmons	ANA Media	
1002		Angela Bassett	Modern VideoFilm	
1003		Ryan Corr	Sony Pictures Worldwide Acquisitions (SPWA)	
1004		Alexander SkarsgÃ¥rd	DJ Films	
1005		Gus Van Sant	Telefilm Canada	
1006		Christopher Lloyd	Glendale Picture Company	
1007		Christopher Walken	Scooter Braun Films	
1008		Devon Aoki	LG	

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1009		Lauren Graham	New Legacy	
1010		Gong Li	Heritage Films	
1011		Garry Marshall	Alberto Grimaldi Productions	
1012		Goran Visnjic	Indelible Productions	
1013		Helena Bonham Carter	FilmfÃ¶rderung Hamburg	
1014		Kevin Costner	World Wrestling Entertainment (WWE)	
1015		Tyrese Gibson	Vif Babelsberger Filmproduktion GmbH Co. Zweite KG	
1016		Derek Richardson	MFP Munich Film Partners GmbH Company I. Produktions KG	
1017		Jeff Daniels	Stephen J. Cannell Productions	
1018		TÃ©a Leoni	Fuzzy Door Productions	
1019		Freddy RodrÃguez	Apaches Entertainment	
1020		Julie Delpy	Ascension Pictures	
1021		Armin Rohde	Smokehouse Pictures	
1022		Allie Mickelson	American Empirical Pictures	
1023		Emilie de Ravin	CAMERON MACKINTOSH	
1024		Eugene Hutz	First Light Production	
1025		Kevin Zegers	New Holland Pictures	
1026		Barry Watson	Yucaipa Films	
1027		Adam Pascal	CinÃ©+	
1028		Marc Devigne	A Hot Dog	
1029		Elle Peterson	Onda Entertainment	
1030		Jerry Mofokeng	British Film Institute (BFI)	
1031		Paul Newman	Partnership Pictures	
1032		Audrey Tautou	Also Known As Pictures	
1033		Ken Davitian	VideoFilmes	
1034		Simon Chandler	Panay Films	
1035		Sean Bean	Cinereach	
1036		Burt Young	Park Pictures	
1037		Jim Belushi	Rai Cinema	
1038		Jeremy Irons	Subterranean Productions LLC	
1039		Kathleen Quinlan	Super Cool ManChu	
1040		Ray Liotta	Disney Enterprises	
1041		Jason Schwartzman	Shady Acres Entertainment	
1042		Ana de la Reguera	Joseph Nittolo Entertainment	
1043		Shawn Wayans	Camelot Pictures	
1044		Julia Stiles	Mediana Productions Filmgesellschaft	
1045		Kazunari Ninomiya	Herold Productions	
1046		Maggie Gyllenhaal	FidÃ©litÃ© Productions	
1047		Sung Kang	Earthship Productions	

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1048		Ian Somerhalder	Parkes+MacDonald Image Nation	
1049		Kristin Davis	Be Gentlemen Limited Partnership	
1050		Joanna 'JoJo' Levesque	Woods Entertainment	
1051		Red Madrell	Esperanto Filmoj	
1052		Michael Sheen	EMJAG Productions	
1053		Cameron Bright	Disney Double Dare You (DDY)	
1054		Amber Tamblyn	Riddick Canada Productions	
1055		Kyle Gass	Emmett/Furla Films	
1056		Abhishek Bachchan	8:38 Productions	
1057		Carrie-Anne Moss	Unique Features	
1058		James Corden	Pop Films	
1059		Billy West	Troika Pictures	
1060		Stephen Graham	Verisimilitude	
1061		Samy Naceri	De Milo	
1062		Holly Hunter	Weinstein Company, The	
1063		Bridget Moynahan	Universal	
1064		Selma Blair	Regione Lazio	
1065		Michael Jeter	Milk Media	
1066		Teri Polo	Five Star Institute	
1067		Jennifer Coolidge	Team G	
1068		Raoul Bova	Emu Creek Pictures	
1069		Cyril Raffaelli	Post Empire Films	
1070		Ving Rhames	U-Drive Productions	
1071		Christina Cox	Parkes/MacDonald Productions	
1072		David Sullivan	Lonely Film Productions GmbH Co. KG.	
1073		Sophie Okonedo	Blue Yonder Films	
1074		Matt Stone	Hart-Sharp Entertainment	
1075		Maia Morgenstern	Patalex V Productions Limited	
1076		Kerry Washington	JC 23 Entertainment	
1077		Oliver James	Clubdeal	
1078		Jonny Lee Miller	Five Roses	
1079		Anthony Anderson	Baldwin Entertainment Group	
1080		Scott Mechlowicz	Eagle Vision Inc.	
1081		Ian Holm	Max Stronghold Productions Inc.	
1082		Alice Braga	UK Film TV Production Company	
1083		Marcia Gay Harden	Second Mate Productions	
1084		Charlie Cox	Skylark Productions	
1085		ZoÃ Bell	Hit Run Productions	
1086		Lena Headey	Konami	

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1087		Eva Mendes	Muse Entertainment Enterprises	
1088		Michael PeÃ±a	C.O.R.E. Feature Animation	
1089		Virginia Madsen	Sound for Film	
1090		Julie Kavner	Craven-Maddalena Films	
1091		Reiko Aylesworth	Pricel	
1092		Melissa George	Black White Productions	
1093		MarkÃ©ta IrglovÃ¡	Iron Film	
1094		Sarah Roemer	MMCB Film Produktion 2004	
1095		Keri Russell	Big Beach Films	
1096		Micah Sloat	Detour Filmproduction	
1097		Scott Patterson	TMC Films	
1098		Monica Bellucci	Sunset Gower Studios	
1099		Dougray Scott	Zero Gravity Management	
1100		Madison Pettis	Anagram Pictures Inc.	
1101		Julia Ormond	Screen Yorkshire	
1102		Brad Dourif	Out of the Blue... Entertainment	
1103		Tim Roth	Sommers Company, The	
1104		Tamala Jones	Universal CGI	
1105		Oded Fehr	Shawn Danielle Productions Ltd.	
1106		Dan Futterman	Riff Raff Film Productions	
1107		David Morrissey	Bob Yari Productions	
1108		Brian Cox	Dylan Sellers Productions	
1109		Scott Glenn	Loofah Productions	
1110		Cristian Solimeno	Valhalla Motion Pictures	
1111		Atossa Leoni	Metropolitan Filmexport	
1112		Juliette Binoche	Primal Foe Productions	
1113		Emily Mortimer	Sideways Productions Inc.	
1114		Jenna Fischer	Friday Night Lights LLC	
1115		Benicio del Toro	John Wells Productions	
1116		Unax Ugalde	Ludlum Entertainment	
1117		Constantino Romero	Lauren Levine Productions Inc.	
1118		Donnie Wahlberg	Grosvenor Park Impact Productions	
1119		Bill Milner	Columbus 81 Productions	
1120		Jack Thompson	Andalasia Productions	
1121		Aaran Thomas	Playtone Production	
1122		Moazzam Begg	Depth of Field	
1123		Grace Thorsen	RTE	
1124		Tom Skerritt	Southpaw Entertainment	
1125		Lois Chiles	Anka Film	

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Master Index	Genre	Actor	Production Company	Director
1126		A. Michael Baldwin	Monkey Dance Productions	
1127		Michael Biehn	Beech Hill Films	
1128		Kate Capshaw	Two Ton Films	
1129		Karen Allen	Nightfall Productions	
1130		Maud Adams	Bueprint Pictures	
1131		Dabney Coleman	Stable Way Entertainment	
1132		Karan Ashley	Tollin/Robbins Productions	
1133		Chris Tucker	MID Foundation	
1134		Sharon Stone	Kernos Filmproduktionsgesellschaft Company	
1135		Jaye Davidson	Jon Shestack Productions	
1136		Daryl Hannah	Lars Productions	
1137		Carole Bouquet	Sunlight Productions	
1138		Lee Van Cleef	Code Entertainment	
1139		Claire Bloom	Evolution Entertainment	
1140		Frances McDormand	Quinta Communications	
1141		Wayne Knight	Hemdale Film	
1142		James Caan	Black Rhino Productions	
1143		Quentin Tarantino	Industrial Light Magic (ILM)	
1144		Stacy Keach	Woodcote	
1145		Joanna Lumley	Cecchi Gori Pictures	
1146		Taylor Momsen	Palace Pictures	
1147		Albert Finney	International Film Investors	
1148		Jesse Bradford	Courtside Seats Productions	
1149		Robin Tunney	Los Hooligans Productions	
1150		Spencer Breslin	Matchmaker Films	
1151		Scott Caan	Munich Film Partners Company (MFP) MI2 Productions	
1152		Lori Heuring	Dino De Laurentiis Company	
1153		Keith David	Technical Black	
1154		Stephen Dorff	Imaginary Forces	
1155		Julia Louis-Dreyfus	Savoy Pictures	
1156		Edward Furlong	Baltimore Pictures	
1157		Mark Margolis	Harvest Filmworks	
1158		Dexter Fletcher	Silver Screen Partners IV	
1159		Jason Scott Lee	Laredo	
1160		Carey Lowell	Larry Franco Productions	
1161		Alex Winter	Starway International Inc.	
1162		Robby Benson	National Film Trustee Company	
1163		Don Johnson	Maljack Productions	
1164		Reggie Bannister	Stage III Productions	

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1165		Patrick Swayze		
1166		Rick Moranis		
1167		Maryam d'Abo		
1168		Lisa Blount		
1169		Christopher Lee		
1170		Robert Shaw		
1171		John Cleese		
1172		Ursula Andress		
1173		Honor Blackman		
1174		Paul McCartney		
1175		Jill St. John		
1176		Harry Guardino		
1177		Joe Pesci		
1178		Liselotte Pulver		
1179		Laura Dern		
1180		Tupac Shakur		
1181		Akiko Wakabayashi		
1182		Daniela Bianchi		
1183		Max von Sydow		
1184		Yaphet Kotto		
1185		Claudine Auger		
1186		Diana Rigg		
1187		Tim Matheson		
1188		Hugh Keays-Byrne		
1189		Carrie Fisher		
1190		Will Poulter		
1191		Emilia Clarke		
1192		Richard Kind		
1193		LÃ©a Seydoux		
1194		Peter Dinklage		
1195		Jennifer Jason Leigh		
1196		Evangeline Lilly		
1197		Raffey Cassidy		
1198		Jennifer Ehle		
1199		Thomas Brodie-Sangster		
1200		Ninja		
1201		Odeya Rush		
1202		Jeffrey Wright		
1203		Emory Cohen		

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Master Index	Genre	Actor	Production Company	Director
1204		Jason Mitchell		
1205		Danila Kozlovskiy		
1206		Kyle Chandler		
1207		Teresa Palmer		
1208		Lily James		
1209		Natalie Martinez		
1210		Tang Wei		
1211		Kate Dickie		
1212		Rodrigo Santoro		
1213		Hannah Ware		
1214		Cillian Murphy		
1215		Graham McTavish		
1216		Christopher Eccleston		
1217		Venus Schultheis		
1218		Gugu Mbatha-Raw		
1219		Ana de Armas		
1220		Charlotte Le Bon		
1221		Adam Rayner		
1222		Joely Richardson		
1223		Linda Cardellini		
1224		Caroline Goodall		
1225		Bella Thorne		
1226		Tate Ellington		
1227		Ian Nelson		
1228		Joey Morgan		
1229		Kathryn Hahn		
1230		Allison Tolman		
1231		Christina Hendricks		
1232		Tony Revolori		
1233		Tatiana Maslany		
1234		Sofia Black-D'Elia		
1235		Loan Chabanol		
1236		Rodger Bumpass		
1237		Lin Shaye		
1238		Frances de la Tour		
1239		Beth Behrs		
1240		Donald Glover		
1241		Pfeifer Brown		
1242		Aurora Perrineau		

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1243		Karen Abercrombie		
1244		Dave Bautista		
1245		Alfie Allen		
1246		Richard Armitage		
1247		Daniel Henney		
1248		Carrie Coon		
1249		Choi Min-sik		
1250		William Fichtner		
1251		Melissa Benoist		
1252		Kyle Gallner		
1253		Elizabeth Olsen		
1254		Craig Berry		
1255		Ben Barnes		
1256		Dave Franco		
1257		Bobby Cannavale		
1258		Lea Thompson		
1259		Stephen Boss		
1260		Olivia Cooke		
1261		Leslie Bibb		
1262		Edwin Hodge		
1263		Eric Ladin		
1264		Daniel Henshall		
1265		Ted Levine		
1266		Enrico Colantoni		
1267		MÃ©lanie Thierry		
1268		Aaron Paul		
1269		Nathan Gamble		
1270		Connor Corum		
1271		Mary Steenburgen		
1272		Patricia Heaton		
1273		David A.R. White		
1274		Curd JÃ¼rgens		
1275		Jordan Nagai		
1276		Nathalie Boltt		
1277		Ayelet Zurer		
1278		Kathy Bates		
1279		Eric Winter		
1280		Jennifer Saunders		
1281		Matthew Perry		

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Master Index	Genre	Actor	Production Company	Director
1282		Steve Valentine		
1283		Dwight Yoakam		
1284		Lorna Raver		
1285		Chandler Canterbury		
1286		Dominique McElligott		
1287		Luis GuzmÃ¡n		
1288		Hugh Laurie		
1289		Giovanni Ribisi		
1290		Nick Zano		
1291		Bryan Greenberg		
1292		Joel McHale		
1293		Ian McShane		
1294		Rhys Ifans		
1295		Judy Greer		
1296		Betsy Russell		
1297		Brian J. White		
1298		Madhavan		
1299		Gaelan Connell		
1300		Lauren Ambrose		
1301		Tony Kgoroge		
1302		Shirley Henderson		
1303		Dervla Kirwan		
1304		Kerr Smith		
1305		Peter Sarsgaard		
1306		Armin Mueller-Stahl		
1307		Jordana Brewster		
1308		William Black		
1309		Tyler Mane		
1310		Emily Osment		
1311		Kevin Nealon		
1312		Andrea Roth		
1313		Connie Britton		
1314		Hakeem Kae-Kazim		
1315		Eddie Marsan		
1316		Paul Schneider		
1317		Martin Starr		
1318		Jena Malone		
1319		Alexis Georgoulis		
1320		Ali Larter		

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Master Index	Genre	Actor	Production Company	Director
1321		Klaus Maria Brandauer		
1322		Alice Krige		
1323		Kathleen Munroe		
1324		Danny Huston		
1325		Tommy Davidson		
1326		Mahesh Manjrekar		
1327		Daryl Sabara		
1328		Rhona Mitra		
1329		Emi Ikehata		
1330		Wen Bin Huang		
1331		Vincent Cassel		
1332		Dolph Lundgren		
1333		Craig Ferguson		
1334		Donna Murphy		
1335		Ned Beatty		
1336		Corey Stoll		
1337		Amanda Bynes		
1338		Kieran Culkin		
1339		Maria Bello		
1340		Jackson Rathbone		
1341		Barbara Hershey		
1342		Joe Slaughter		
1343		Oliver Platt		
1344		DragoÈ™ Bucur		
1345		Mike Vogel		
1346		Gio Perez		
1347		Danneel Ackles		
1348		Bailee Madison		
1349		Kasia Smutniak		
1350		Brian Dennehy		
1351		Jenn Proske		
1352		Barack Obama		
1353		Tom Selleck		
1354		Kevin Durand		
1355		Richard Jenkins		
1356		Anthony Edwards		
1357		Eric Dane		
1358		Chris Cooper		
1359		Brian Boland		

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1360		Juan Carlos HernÃ¡ndez		
1361		Annalee Jefferies		
1362		Jeffrey DeMunn		
1363		Jerry O'Connell		
1364		Iris Bahr		
1365		Ty Burrell		
1366		Robert Paterson		
1367		Stephanie Szostak		
1368		David Zayas		
1369		Kevin Breznahan		
1370		Val Kilmer		
1371		Stephen Spinella		
1372		Anna Friel		
1373		Glenn Howerton		
1374		Dennis Hopper		
1375		Tim McGraw		
1376		Phoebe Tonkin		
1377		Bojana NovakoviÄ‡		
1378		Ariel Schulman		
1379		Brooke Shields		
1380		Hannah Endicott-Douglas		
1381		Sam Huntington		
1382		Mary-Louise Parker		
1383		Amancaya Aguilar		
1384		MoisÃ©s Arias		
1385		Nick Damici		
1386		Kayvan Novak		
1387		Taika Waititi		
1388		Meat Loaf		
1389		John Hannah		
1390		David Morse		
1391		Robert Carlyle		
1392		Kelsey Grammer		
1393		Gina McKee		
1394		Madison Eginton		
1395		Kevin Pollak		
1396		Delroy Lindo		
1397		Joshua Leonard		
1398		Linda Fiorentino		

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Master Index	Genre	Actor	Production Company	Director
1399		Bob Zmuda		
1400		Hart Bochner		
1401		Mary Gibbs		
1402		Claudia Christian		
1403		Alexander Pollock		
1404		Mary McCormack		
1405		Jeremy Sisto		
1406		Isaiah Washington		
1407		Obba BabatundÃ©		
1408		Michael Showalter		
1409		Jeff Garlin		
1410		Mathieu Amalric		
1411		Clark Duke		
1412		Bee Vang		
1413		David Schwimmer		
1414		Anil Kapoor		
1415		Brian Geraghty		
1416		Matthew Marsden		
1417		Sarah Jane Morris		
1418		Odette Annable		
1419		Susie Essman		
1420		Donald Sutherland		
1421		Beau Bridges		
1422		Jennifer Schwalbach Smith		
1423		Kat Dennings		
1424		Neal McDonough		
1425		Emmanuelle Chriqui		
1426		Colin Hanks		
1427		Alex Fisher		
1428		Mary Kay Place		
1429		Toby Kebbell		
1430		Parker Posey		
1431		Charlie Yeung		
1432		Cynthia Nixon		
1433		Jennifer Hudson		
1434		name		
1435		Hayley Atwell		
1436		Gabrielle Union		
1437		Sergei Bezrukov		

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Master Index	Genre	Actor	Production Company	Director
1438		Nicole Ari Parker		
1439		Ellen Burstyn		
1440		Yorick van Wageningen		
1441		Bill Pullman		
1442		Luis Fernando PeÃ±a		
1443		William Dembski		
1444		Charlie Day		
1445		Patricia Clarkson		
1446		Jared Harris		
1447		Greta Gerwig		
1448		Maya Rudolph		
1449		Riley Griffiths		
1450		Violante Placido		
1451		Alfred Molina		
1452		January Jones		
1453		John Slattery		
1454		Isabel Lucas		
1455		Callum Blue		
1456		Will Rothhaar		
1457		Miles Fisher		
1458		Justin Michael Brandt		
1459		Ulrich Thomsen		
1460		Kenneth Branagh		
1461		Octavia Spencer		
1462		Frank Grillo		
1463		Robin Wright		
1464		Rachel Nichols		
1465		Gillian Anderson		
1466		Mary-Kate Olsen		
1467		Elias Koteas		
1468		Lloyd Owen		
1469		Pink		
1470		Courteney Cox		
1471		Dan Fogler		
1472		John Boyega		
1473		Vicky Krieps		
1474		Tim Blake Nelson		
1475		Aidan Gillen		
1476		Amara Miller		

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1477		Kaley Cuoco		
1478		Liam Cunningham		
1479		Lauren Bittner		
1480		Liana Liberato		
1481		James D'Arcy		
1482		Ryo Kase		
1483		Alice Parkinson		
1484		Sarah Shahi		
1485		Lorraine Nicholson		
1486		Aziz Ansari		
1487		Jessica Szohr		
1488		Katy Mixon		
1489		Nicholas Braun		
1490		Ruth Amarante		
1491		Ben Davies		
1492		Brandon T. Jackson		
1493		April Bowlby		
1494		Fan Bingbing		
1495		Branko ÄuriÄ‡		
1496		Meagan Good		
1497		Amy Wren		
1498		Rosemarie DeWitt		
1499		Amy Ryan		
1500		Rachel Bilson		
1501		Harry Lloyd		
1502		Joel David Moore		
1503		Robb Wells		
1504		Michael O'Keefe		
1505		Jonathan Race		
1506		Usher Raymond		
1507		Tyler Dawson		
1508		Caryn Waechter		
1509		Denis Leary		
1510		Eric Mabius		
1511		Rip Torn		
1512		Frank Finlay		
1513		Brittany Murphy		
1514		Peter Coyote		
1515		George W. Bush		

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1516		Phyllida Law		
1517		James Earl Jones		
1518		Jason Mewes		
1519		James Fleet		
1520		Sarah Peirse		
1521		Jean-Pierre Cassel		
1522		Michael Ealy		
1523		Ayush Tandon		
1524		Chris Wedge		
1525		Seth MacFarlane		
1526		Tom Holland		
1527		Jane Lynch		
1528		Brittany Snow		
1529		Ezra Miller		
1530		Alan Arkin		
1531		Olivia Munn		
1532		Anna Hutchison		
1533		James Hong		
1534		Jared Gilman		
1535		Nestor Serrano		
1536		David Spade		
1537		Aasif Mandvi		
1538		GÃ¶tz Otto		
1539		Diego Boneta		
1540		Jeremy Piven		
1541		Joseph Gilgun		
1542		Justin Theroux		
1543		Sidney Johnston		
1544		Catherine O'Hara		
1545		Eva Amurri Martino		
1546		Katrina Bowden		
1547		Ben Mendelsohn		
1548		Chris Diamantopoulos		
1549		Debbie Reynolds		
1550		Jeff Daniel Phillips		
1551		Dania Ramirez		
1552		Sam Riley		
1553		Evan Helmuth		
1554		Tom Arnold		

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Master Index	Genre	Actor	Production Company	Director
1555		Levy Easterly		
1556		Tom Felton		
1557		Shannon Woodward		
1558		Vivica A. Fox		
1559		Alexander Gould		
1560		Philip Baker Hall		
1561		Brett Kelly		
1562		Mike White		
1563		Charles S. Dutton		
1564		Bruce Dern		
1565		Justin Bartha		
1566		Clint Howard		
1567		Peter Mullan		
1568		Charles Dance		
1569		John Alderton		
1570		Paul Sorvino		
1571		James Cameron		
1572		Frances Fisher		
1573		Rya Kihlstedt		
1574		Jonathan Groff		
1575		Benjamin Bratt		
1576		Katee Sackhoff		
1577		Ron Livingston		
1578		Sebastian Koch		
1579		Abigail Hargrove		
1580		Douglas Smith		
1581		Famke Janssen		
1582		Kevin Bacon		
1583		Morris Chestnut		
1584		Shiloh Fernandez		
1585		Max Martini		
1586		Vicky McClure		
1587		Megan Charpentier		
1588		Ricky Gervais		
1589		Nick Swardson		
1590		Sylvia Hoeks		
1591		Lee Tergesen		
1592		Rila Fukushima		
1593		Michaela Watkins		

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1594		Shannon Ashlyn		
1595		James Deen		
1596		Tiya Sircar		
1597		Tanya Roberts		
1598		Djimon Hounsou		
1599		Faith Ford		
1600		Youki Kudoh		
1601		Don Knotts		
1602		AdriÃ¡n Alonso Barona		
1603		Will Yun Lee		
1604		Emily Watson		
1605		Shirley MacLaine		
1606		AndrÃ© Benjamin		
1607		Eythor Gudjonsson		
1608		Jessica Simpson		
1609		Alex Borstein		
1610		Bob Hoskins		
1611		Craig Archibald		
1612		Sandra Oh		
1613		Nora Zehetner		
1614		Boris Leskin		
1615		Elizabeth PeÃ±a		
1616		Skye McCole Bartusiak		
1617		Chad Faust		
1618		Thomas Curtis		
1619		Terry Pheto		
1620		Tony Curran		
1621		Bonnie Hunt		
1622		Mads Mikkelsen		
1623		Luenell		
1624		David Calder		
1625		Laurie Holden		
1626		Milo Ventimiglia		
1627		Eddie Izzard		
1628		Shohreh Aghdashloo		
1629		Vinessa Shaw		
1630		Joseph Ruskin		
1631		Gary Cole		
1632		Judy Davis		

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1633		Edie Falco		
1634		HÃ©ctor JimÃ©nez		
1635		Seamus Davey-Fitzpatrick		
1636		Tsuyoshi Ihara		
1637		Gary Sinise		
1638		Brian Tee		
1639		Christina Milian		
1640		Danny Glover		
1641		Woody Allen		
1642		Noel Clarke		
1643		Kevin Smith		
1644		Edison Chen		
1645		Ronnie James Dio		
1646		Tommy Flanagan		
1647		Aishwarya Rai Bachchan		
1648		Dylan Baker		
1649		Shannen Fields		
1650		Stephen Campbell Moore		
1651		Ronald Reagan		
1652		Jo Hartley		
1653		Roschdy Zem		
1654		Gena Rowlands		
1655		Tim Meadows		
1656		Rupert Evans		
1657		Eddie Deezen		
1658		Emmy Rossum		
1659		Dominic Purcell		
1660		Ewen Bremner		
1661		Tony D'Amario		
1662		Jake Weber		
1663		Colm Feore		
1664		Vernon Dobtcheff		
1665		Casey Gooden		
1666		Kristen Miller		
1667		Christo Jivkov		
1668		Regina King		
1669		David Keith		
1670		Kathryn Morris		
1671		Hannah Spearritt		

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1672		Trevor Morgan		
1673		Janet Leigh		
1674		Lou Romano		
1675		William Hurt		
1676		Wes Bentley		
1677		Christopher Mintz-Plasse		
1678		Cameron Richardson		
1679		Nancy Cartwright		
1680		John Ortiz		
1681		Dakota Blue Richards		
1682		Hugh Walsh		
1683		Nicky Katt		
1684		Mark Fredrichs		
1685		Kyra Sedgwick		
1686		Hank Azaria		
1687		Daeg Faerch		
1688		Michael Pitt		
1689		Paul Rae		
1690		Sam Shepard		
1691		Terry Crews		
1692		Adam Brody		
1693		Imelda Staunton		
1694		Adam James		
1695		Shaun Toub		
1696		Bob Gunton		
1697		Jessica Hynes		
1698		Christopher Beiring		
1699		Ted Raimi		
1700		Veronica Cartwright		
1701		Michael Lonsdale		
1702		Bill Thornbury		
1703		Linda Hamilton		
1704		Jonathan Ke Quan		
1705		Charles Martin Smith		
1706		Louis Jourdan		
1707		Don Rickles		
1708		Chazz Palminteri		
1709		Izabella Scorupco		
1710		Julian Sands		

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1711		Johnny Yong Bosch		
1712		Nia Long		
1713		Tom Sizemore		
1714		Linda Larkin		
1715		George Dzundza		
1716		Miranda Richardson		
1717		Paul Freeman		
1718		Chaim Topol		
1719		Pearl Bailey		
1720		Ernest Borgnine		
1721		Charles Cyphers		
1722		Emmanuelle BÃ©art		
1723		Vanessa Williams		
1724		Alan Cumming		
1725		Joe Pantoliano		
1726		Jeffrey Tambor		
1727		Cole Hauser		
1728		Ryan Hurst		
1729		Huntley Ritter		
1730		Kristen Johnston		
1731		Matthew Settle		
1732		Sean Young		
1733		Robert MacNaughton		
1734		Wilford Brimley		
1735		Beverly D'Angelo		
1736		Claire Forlani		
1737		Ben Shenkman		
1738		Katie Sagona		
1739		Nick Moran		
1740		Jason Isaacs		
1741		Denholm Elliott		
1742		Robert Davi		
1743		George Carlin		
1744		Richard White		
1745		Chelsea Field		
1746		Meg Foster		
1747		Jerry Orbach		
1748		Jeroen KrabbÃ©		
1749		Victor Wong		

Kaggle Key 2 continued from previous page

Master Index	Genre	Actor	Production Company	Director
1750		Britt Ekland		
1751		Eric Idle		
1752		Barry Bostwick		
1753		Joseph Wiseman		
1754		Gert FrÃ¶be		
1755		George Harrison		
1756		Charles Gray		
1757		Reni Santoni		
1758		Don Pedro Colley		
1759		Cybill Shepherd		
1760		Daniel Stern		
1761		Horst Buchholz		
1762		Juliette Lewis		
1763		Karin Dor		
1764		Lotte Lenya		
1765		Dennis Dun		
1766		Jane Seymour		
1767		Adolfo Celi		
1768		Telly Savalas		
1769		P.J. Soles		
1770		John Vernon		

Table 3: Genre Key (Both Sets)

Master Index	Genre	MATH512 Genre
0	Action	Adventure
1	Adventure	Action
2	Western	Thriller
3	Science Fiction	Drama
4	Drama	Comedy
5	Family	Musical
6	Comedy	Western
7	Crime	Horror
8	War	Rom-Com
9	Fantasy	Thriller/Suspense
10	Thriller	Black Comedy
11	Romance	Concert/Performance
12	Mystery	
13	Horror	
14	History	
15	Animation	
16	Documentary	
17	Music	
18	Foreign	
19	blank	

Table 4: Top Market Key (*thenumbers* Set)

Master Index	Top Market Name	Frequency
0	United Kingdom	354
1	Japan	198
2	China	708
3	South Korea	193
4	Mexico	78
5	India	43
6	Russia	95
7	Spain	47
8	France	146
9	Australia	97
10	Italy	40
11	Germany	68
12	Saudi Arabia	2
13	Netherlands	4
14	Turkey	14
15	Poland	17
16	Brazil	32
17	Indonesia	2
18	Czech Republic	1
19	Venezuela	6
20	Taiwan	4
21	United Arab Emirates	3
22	Argentina	3
23	Chile	1
24	New Zealand	1
25	Egypt	2
26	Hong Kong	2
27	Ukraine	2
28	Bulgaria	1
29	South Africa	1
30	Malaysia	1
31	Belgium	1
32	Singapore	2
33	Portugal	1

11.2 Entire Graphs

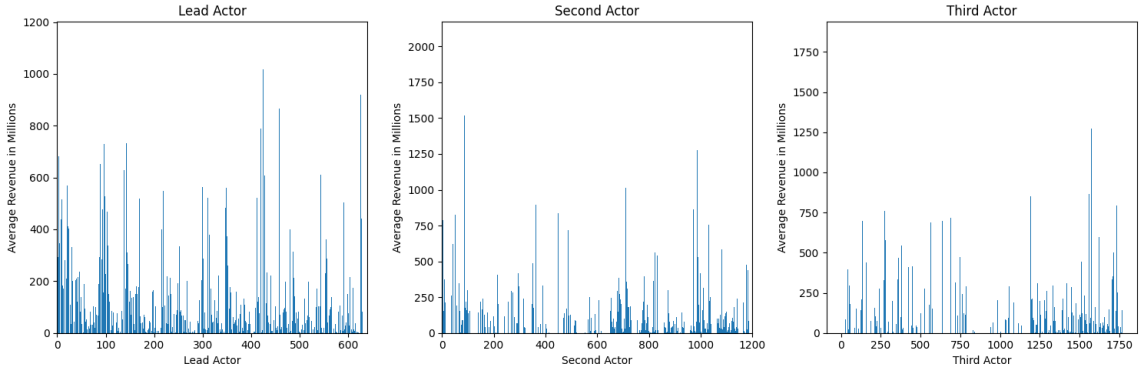

Figure 27: Plot of Average Revenues (millions) vs Actors from *Kaggle* data set

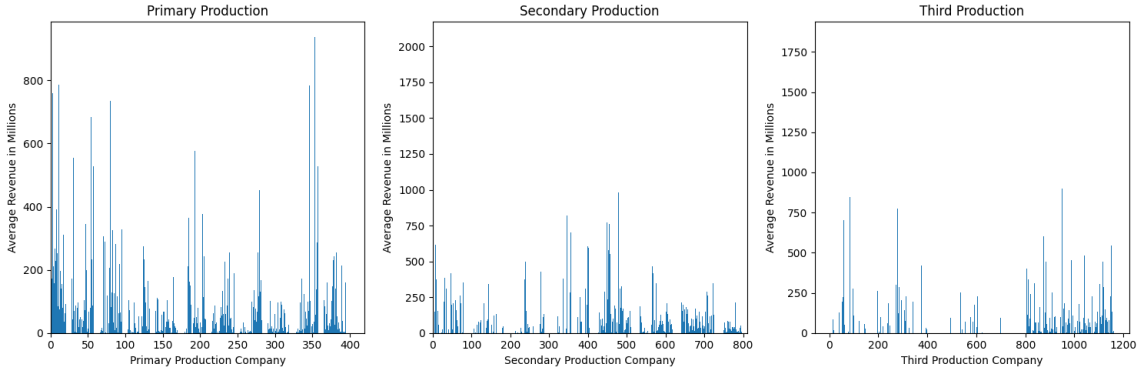

Figure 28: Plot of Average Revenues (millions) vs Production Company from *Kaggle* data set

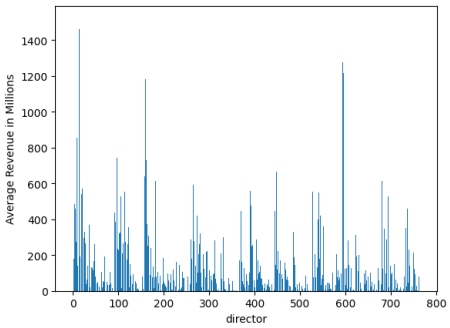

Figure 29: Plot of Average Revenues (millions) vs Director from *Kaggle* data set

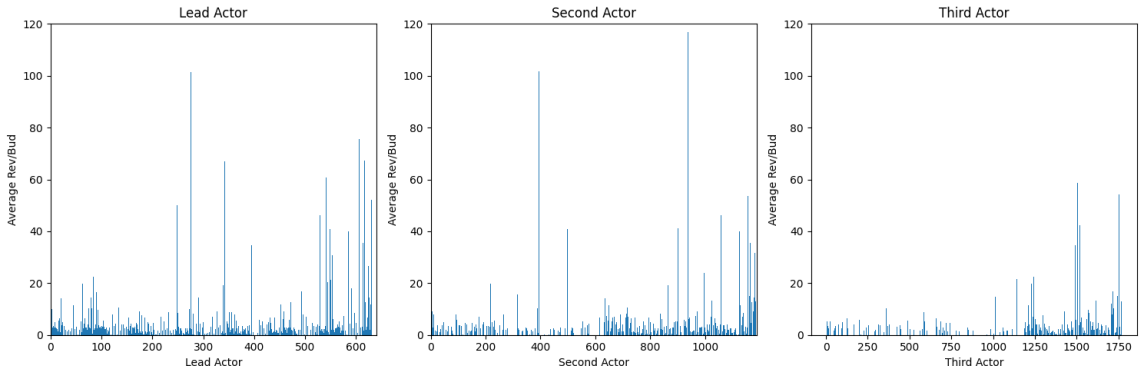

Figure 30: Plot of Average Rev/Bud vs Actors from *Kaggle* data set

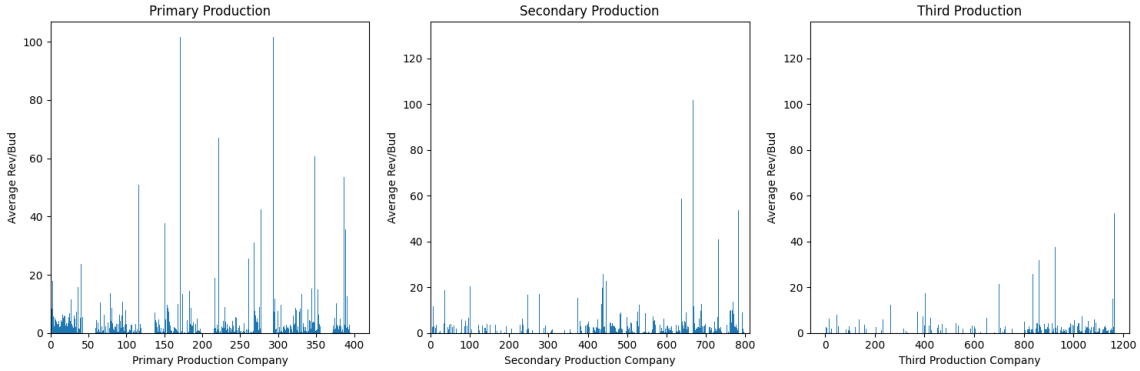

Figure 31: Plot of Average Rev/Bud vs Production Companies from *Kaggle* data set

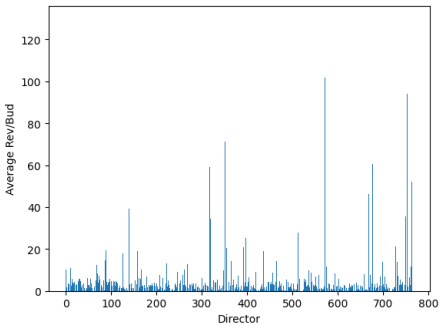

Figure 32: Plot of Average Rev/Bud vs Directors from *Kaggle* data set